

SZABIST
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Real-Time Air Quality Monitor and Predictor

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Real-Time Air Quality Monitor and Predictor

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DECLARATION

We, the candidates of Bachelor of Science (Computer Science) at Shaheed Zulfikar Ali Bhutto Institute of Science and Technology, Islamabad do hereby certify that this report titled **Real-Time Air Quality Monitor and Predictor**, submitted as partial fulfillment of Bachelor of Science (Computer Science) degree requirements, is our original work and we are its sole author. All the employed materials, references to the literature and the work of others have been referred to and duly cited. This report has not been presented for examination anywhere else.

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Project Overview

The proposed system aims to design and develop a scalable platform capable to monitor, analyze, and predict levels of air quality in real-time and in different locations. The solution is expected to provide users with live dashboards, visual data displays of the past, and present air conditions, as well as forecast pollution. The air quality prediction system equipped with AI algorithms to ensure accuracy. The proposed product collects data to help local authorities in rural and urban areas in managing the environment, improving the level of air quality, and helping in making informed policy decisions.

The selected proposal would create a prototype of a system that will be able to scale up to monitor pollution. The system will have IoT sensors for air pollutants and will use AI to make predictions regarding pollution. The system will use APIs to make it more accurate and provide the ability to offer health warnings based on the predictions and the users' locations. The web application will be easy to use and provide access to the information visually.

The objective will be to design and implement a scalable system for monitoring, analyzing, and predicting air quality in various locations. The system will utilize IoT sensors to detect pollutants and apply artificial intelligence technology to predict and track changes in air quality. The proposed system will also use external APIs to enhance performance and issue location-based health warnings to the system users. The system will have a web and mobile interface that allows the general public to access air quality information.

The proposed system will be further improved. It will be possible to increase the amount of sensors to improve the accuracy of making predictions using deep learning algorithms. It will be integrated into the infrastructure of smart cities. The system will be developed to accept feedback from users, by providing crowdsourcing services for environmental monitoring. The system will be made more convenient for use and compatible with other platforms. These changes will allow the system to better address the problem of air pollution.

Dedication

I dedicate my desire for knowledge and excellence to divine blessings. I am grateful to the most honorable Prophet Muhammad (peace be upon him) whose teachings made me thirsty for knowledge for the sake of Allah and his religion. I would like to express my deepest gratitude to my Mentors especially Sir Ibn e Amin for constantly encouraging me and supporting me to achieve my goals.

The work is dedicated to my family for their support and understanding, without which I would have not succeeded. To my family, to all family members, I am devoted, and my respect is endless. To my teachers, whose advice, support, and knowledge gave me an opportunity to get an education. To the representative of the organization, for your advice and support during the work on the work.

Each acknowledgment highlights significant roles played by individuals who helped achieve the outcomes. The family members have supported unconditionally and believed in the dreams and vision. Their encouragement always played a role in overcoming challenges. The teachers offered guidance, which enabled one to gain knowledge and wisdom. Their mentorship helped improve skills and boosted critical thinking. Every individual involved in the process made outstanding efforts and played a crucial role. The support and encouragement they provided will always be appreciated and cherished.

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Chapter 1

Introduction

Air pollution monitoring and prediction systems offer an effective method of tracking and understanding air quality. Sensors are used in these systems to measure harmful pollutants such as PM2.5, NO, CO etc. while intelligent prediction models analyze collected data on air quality and other environmental factors to estimate future conditions of air quality. The system combines monitoring and predictive capabilities to provide users with a clearer understanding of changing air conditions. Additionally, the system includes several useful features. These include, but are not limited to, the ability to track pollution trends, review historical records, and generate alerts at specific locations when air quality reaches unhealthy levels. The system's flexible design allows it to be deployed in urban areas as well as remote sites, making it suitable for a wide range of environments. By providing easy access to accurate and timely information regarding air quality, individuals, communities, and decision makers can make informed decisions concerning health and environmental protection. The predictive system provides accurate results for better output. A front-end design is also provided that makes the system easily understood by users. Through user-centered design and advanced technology, the system supports cleaner air; raises public awareness; and contributes to healthier living conditions.

1.1 Product Purpose

The device is created to monitor the quality of the air all the time, predict pollution through internet connected sensors and smart computing. The device collect data from a wide variety of things that are in the air include PM2.5, CO, NOx throughout many locations. The system uses real-time trusted sources (OpenWeather and IQ Air) for accuracy. The computer programs take the data they have collected along with data from other trusted sources to make accurate predictions regarding pollution. Once the predictions are made, the system send health suggestions to people based upon the bad things in the air. The device easily use via web or phone so as to enable use of the device by people, environmental authorities, city planners, and public health officials. It also provides a means for people studying the environment to see what is happening to the air over time which enables these individuals to plan for health and attempt to control pollution. It also allows members of an individual community to know what is occurring which allows them to alter their behavior and allows decision makers to make decisions based on fact. The product wish to reduce things that people breathe, maintain healthy people, support good environments in many different areas by monitoring the air and predicting pollution.

Air pollution is an essential problem for the whole ecosystem, human health, and even the daily routine of every person. Not all the locations are prepared to timely respond to the hazardous changes in the air. The system under consideration addresses the issue by implementing the most innovative approaches that allow monitoring the air quality. It uses particular algorithms to track the level of PM2.5, PM10, CO2, NO2, and other

indicators. Following that, the data is organized into practical suggestions for humans so that they could respond to the current situation. The system relies on OpenWeather and IQAir to ensure the accuracy of the provided information. In addition, the system is designed to offer health advice and alerts so that people could make smart choices and reduce the dangerous impact of pollution on their bodies. The tool also allows analyzing the patterns of behavior in the selected area to contribute to the improvement of the existing policies and regulations. The system's interface is user-friendly and suitable for both regular citizens and scientists. It empowers them to shift from the traditional approaches to a more proactive attitude toward the environmental issues.

Air pollution remains a major challenge in Pakistan and affects public health, environmental quality, and daily life. Reliable monitoring and prediction of air conditions play an important role and provide timely and accurate information. The air quality monitoring and prediction system combines sensor technology with predictive models, detects harmful pollutants, and estimates air quality trends. The system helps individuals, environmental organizations, and government authorities make informed decisions and supports healthier and cleaner communities. The system measures pollutants such as PM2.5, PM10, CO, and NO. These pollutants contribute to respiratory and environmental problems, especially in densely populated cities such as Lahore, Karachi, and Islamabad. The system analyzes air quality data, generates forecasts, provides early awareness of potential pollution events, and supports preventive action. The system contributes to environmental efforts, promotes responsible air quality management, and supports goals that focus on public health, sustainable cities, and climate action. Easy access through digital platforms increases public participation. Information from trusted international sources strengthens decision-making for urban plans, industrial regulation, healthcare services, and environmental policies. Across urban and rural areas makes the system valuable for communities throughout Pakistan. Predictive analysis, and accessible information promote environmental awareness, and contribute to environmental protection across the country.

1.2 Product Scope

The air pollution system is made to watch study and predict air pollution in different places. It uses sensors to measure bad things in the air like PM2.5, PM10, NO, CO and CO. The system gets information from other trusted sources like OpenWeather and IQAir to make its predictions better. The air pollution system uses computer programs to look at the information from the sensors and other sources to make predictions about the air quality. The system has easy to use websites and mobile apps so people like individuals, environmental agencies, healthcare professionals, researchers and urban planners can use it. It has things like live air quality updates, old data to look at and health warnings based on where you're. The air pollution system sends out warnings to help people stay healthy. It works well in cities and in the country no matter what the roads and buildings are like. The air pollution system helps people know about the air and encourages them to be kind to the environment. It also helps people make choices, about the environment and public health. The air pollution system gives people the information they need to control pollution stay healthy and have an environment. The system gets updated regularly. Can be used by more people over time. The air pollution system also lets people work together to watch and improve the air quality in their area. [2].

The system is really good because it uses Internet of Things sensors and gets information from places.The pollution forecasting system and the awareness it brings are very important. The system is also good, for making decisions based on the pollution data it collects.

Module: User Login And Management

- Use Authentication.
- Role-based access control (admin).
- Admin add, delete, view or restrict login access.
- Role-based access control (user).
- Users modify name and password.
- User session management activity logs.

Module: Sensor Data Storage

- Storing Real-time data from sensor.
- View sensor data in real time.
- Automated backup and replication.
- Timestamp-based data indexing for quick retrieval
- API endpoints for adding/updating sensor data
- Sensor Integration between hardware and software

Module: Real-Time Air Quality Data

- Fetch air quality data from APIs (Open Weather)
- Web Socket-based real-time AQI updates
- Convert raw API data into structured JSON
- Historical AQI trend analysis
- Export AQI data in CSV, JSON, or Excel
- Prediction reporting

Module: Live DashBoard

- User Dashborad
- Sensor Values
- Weather Data
- Google Maps/Leaflet.js for sensor location mapping

Module: Data Processing and Cleanup

- Data normalization and error handling.
- Outlier detection using statistical methods.
- Error handling for incomplete sensor readings

- Automated data archiving for old records

Module: Live Sensor Location Tracking

- GPS-based real-time sensor tracking
- WebSockets for live map updates
- Geofencing alerts for predefined zones
- Search filter sensors by location

Module: Alerts and Notifications

- WebSocket-based real-time alerts.
- Email, SMS, and push notifications.
- In-app alert panel for warnings.
- SMS alerts with Twilio API

Module: Report Generation and Export

- DF, CSV, and Excel report generation
- Automated daily/weekly email reports.
- CSV/Excel data export for analysis
- Generate PDF reports with jsPDF

Module: AI/ML Smog Prediction

- Time-series forecasting (LSTM, ARIMA)
- Regression models for pollution prediction
- Anomaly detection using K-means clustering
- Real-time AQI classification using ML models

Module: Health Safety Insights

- Real-time AQI-based health tips
- Location-based air quality warnings
- Weather AQI correlation analysis
- Emergency alerts for hazardous conditions

Module: User Feedback Complaints

- User-friendly complaint submission form
- Image file upload support (pollution evidence)
- Admin dashboard for complaint management
- Secure database storage for user complaints

Module: Offline Mode Dark Mode

- Service Workers for offline data caching
- Indexed DB for storing historical AQI data
- Background Sync for offline updates
- Theme toggle for dark/light mode

1.3 Objectives

The main idea of our air quality monitoring system is to present a solution that offers the creation of an extensive list of predictions that cover the pollution level in different areas. The data collected by IoT sensors that are capable of reading the content of PM2.5, PM10, NO2, CO, and CO2 in the air in real-time. The information from the sensors and the air quality indexes of the area received from external APIs such as OpenWeather and IQAir into forecasts. The latter ones, along with the health recommendations based on the pollution levels, available in web and mobile interfaces that aim to be intuitive and user-friendly. The software allow for individual, local, and global perspectives on air quality, providing online access to personal, community, governmental, and scientific data. It also support the improvement of air quality in the areas of interest and promote responsibility and engagement in sustainability in everyday life through data transparency. The system also ensure smart air quality management for urban and rural areas with high accuracy and sufficient scalability, contributing to better planning of communities in terms of ecosystem management and sustainable development. [1].

The main purpose of the system is to create an intelligent and adaptive solution for controlling, predicting, and observing air pollution levels so people, organizations, and communities can make timely decisions to protect public health and the environment from dangerous levels of contamination. The system helps monitor the spread of air pollution using IoT sensors and artificial intelligence technologies to detect and predict hazardous concentrations of PM2.5, PM10, NO2, CO2, and other pollutants at local and regional levels. In addition, the system provides relevant and up-to-date information on pollution levels based on the user's location, sensor data, and external sources such as OpenWeather and IQAir. Moreover, it offers personalized health advice and recommendations for people at risk based on their health indicators and pollution levels. The system allows users to track present and past pollution levels and observe trends at the street level, helping them take timely actions to reduce exposure to air pollution. The solution is available to people across the globe and is beneficial for mayors, local authorities, medical workers, researchers, and the general public. It helps raise awareness of the impact of air pollution on the environment and health, promotes engagement in addressing environmental challenges, and encourages data-driven decision-making at local and global levels. The proposed concept improves air quality management by making it more accessible and transparent while supporting efforts to reduce pollutant levels across different regions and countries.

The system will be developed with a focused vision: to empower individuals, communities, and institutions with intelligent tools that will transform how air quality is monitored, understood, and managed. Its primary goal will be to protect public health by reducing exposure to harmful air pollutants. By delivering precise, location-based data

and personalized health alerts, the system will enable proactive decisions that promote safer, healthier living—particularly in heavily polluted urban and industrial regions across Pakistan. It will support environmental agencies, health departments, and urban planners in creating effective, data-driven policies. Through advanced pollutant detection, AI-powered forecasting, and long-term trend analysis, the system will provide actionable insights that will shape sustainable urban development and pollution control strategies. The platform will also raise environmental awareness by engaging users through accessible mobile and web applications. It will encourage widespread community participation in air quality tracking and strengthen shared responsibility for environmental well-being. Adaptable to both urban centers and remote rural areas, the system will bridge advanced technology with local environmental needs. By doing so, it will address critical public health concerns, support informed decision-making, and contribute to a cleaner, more resilient, and sustainable future for communities throughout Pakistan.

1.4 Intended Market of Product

The air quality monitoring and prediction system is designed for organizations and professionals responsible for managing and controlling environmental quality and public health within a region. Municipal administrations, health institutions, environmental departments, urban planners, and researchers use the system to monitor air quality, study its impact on health and the environment, and make informed decisions. The tool collects air quality data through Internet-connected sensors and provides access to imake informed decisions regarding policy development, urban planning, and pollution control. In healthcare, the tool supports the study of the effects of air pollution on human health, facilitating the development of appropriate policies and programs. Environmental departments use the system to assess environmental conditions, evaluate the effectiveness of policies and actions, and improve air quality management. Scientific institutions apply the system to conduct research on environmental and social aspects of air quality. The system can be used in both urban and rural areas, making it suitable for organizations requiring flexible environmental management and research solutions. The air quality monitoring and prediction system supports multiple stakeholders in the development and implementation of policies and initiatives for managing air pollution and promoting environmental sustainability in Pakistan.

Target Market: Environmental and public sector organizations.The system will be designed for environmental agencies, municipal administrations, healthcare institutions, urban planning authorities, and research organizations across Pakistan. Data Accuracy

1.5 Intended Users of the Product

The air quality monitoring and prediction system is intended for users and professionals in managing and regulating the quality of the environment and public health. Municipal administrations, health institutions, environmental agencies, urban planners, and researchers supervise air quality, examine its effects on health and the environment, and analyze it to make informed policy and regulatory decisions. Air quality monitoring and prediction tools are used in healthcare to determine the impact of pollution on health and develop appropriate policies. Environmental agencies analyze data to assess the state of the environment, analyze the effectiveness of policies and actions, and develop strate-

gies for controling and reducing emissions. Academic institutions use the tool to carry out analyses of / conduct scientific studies of environmental and social aspects. The system is suitable for both urban and rural areas, making it a viable option for organizations that require a versatile tool for managing and regulating the environment and researching air quality. The air quality monitoring and prediction system help many organizations manage and regulate air quality and develop policies and activities to reduce emissions in Pakistan.

User Role: Environmental Monitoring Agencies. Feature Name

Chapter 2

Background and Literature Review

The literature review is like a foundation for the study. It helps to see what has already been done and tells where to go from here. The literature review of air pollution studies shows that some people use sensors and AI algorithms to keep an eye on the air and predict what it will be like. They make systems that use real time data from things like OpenWeather and IQAir to make guesses, about the air and to help keep people healthy. Air pollution is a problem and the literature review of air pollution helps to understand. [3].

The following sections will look at the air quality monitoring systems that're already in place. How people are going to use these systems and what problems they will be facing. The main problem with air quality monitoring systems and the things that are limiting them will be talked about.

2.1 Existing System Description

Examining existing air quality monitoring systems holds significant value as it uncovers essential knowledge about operational technologies, applied methods, and present limitations. It emphasizes effective features to incorporate and challenges to address, aiding the development of more responsive and precise monitoring solutions. Through such evaluation, developers acquire insights into sensor deployment, environmental data processing, and interface design. The approach promotes informed enhancements, guarantees practical use, and prevents repeated errors. It also reveals successful techniques that improve accuracy, reliability, and user accessibility across diverse air quality tracking and forecasting systems. [4].

The approach reviews five existing air quality monitoring systems, examining their features, technologies, and performance. The review process evaluates each system methodically, with a comparison table offering a clear, detailed analysis to highlight strengths and areas for improvement.

AirGuard is an innovative solution that improves air quality surveillance and forecasting by implementing IoT-based solutions. The technology enables the collection of real-time data and utilizes machine learning algorithms to predict pollution levels. By considering factors such as weather and air quality indexes, AirGuard helps people track changes in the environment. Through the application's smart prediction methods, cities can make informed decisions when addressing air quality issues. Information from global air quality reports can support accurate analysis and evaluation of the situation. The data collected can also help generate insights on the patterns of pollution levels and appropriate measures to control them. People can also receive warnings and alerts about severe pollution levels and guidelines on keeping healthy during such conditions. The use of informative dashboards display-

ing statistical data in graphical formats helps people understand air quality problems better.: undefined [5].

The main features of the Application are listed below:

- Real-Time Air Quality Data: provides live updates on pollutants like PM2.5, CO2, and NO2; AI-Powered Predictions
- AI-Powered Air Quality Prediction: Uses advanced machine learning algorithms to forecast future air quality trends, enabling users to plan outdoor activities safely and reduce exposure to harmful pollutants.
- Interactive Data Visualization Dashboard: Displays real-time air quality information through intuitive charts, graphs, and maps, helping users understand pollution levels easily and make informed environmental decisions.
- Instant Alerts and Notifications: Sends timely alerts about hazardous air quality conditions in the user's area, ensuring they take preventive actions immediately to safeguard their well-being and health.
- Customizable Dashboard Settings: Allows users to personalize their interface, choose preferred pollutant indicators, and manage alert preferences, creating a tailored air monitoring experience that fits their lifestyle best.
- Smart Device Integration: Connects seamlessly with IoT-enabled air sensors, wearables, and smart home devices to provide synchronized, real-time air quality monitoring and automated responses for cleaner living spaces.

The limitations of the application are listed below:

- AirGuard has some limitations: Platform Limitation
- Platform Limitation: Detection may be slower on Android due to Bluetooth restrictions. Location Access
- Platform Limitation: Detection may be slower on Android due to Bluetooth restrictions. Location Access
- Constant Internet Requirement: The app needs a stable internet connection to provide live air quality updates and forecasts. Without active connectivity, users cannot access real-time data or receive alerts effectively.
- Limited Offline Access: The app does not allow offline data viewing or saving past reports, restricting users from checking previously viewed information when they lose internet access.
- Complex Interface for New Users: The app's detailed analytics, pollutant breakdowns, and interactive dashboards may initially seem confusing for beginners unfamiliar with air quality metrics or technical terms.

Clarity Air is an innovative environmental monitoring application created by Clarity Movement Co. which uses Internet of Things or IoT-enabled air quality sensors plus satellite sources to provide accurate information about pollution levels. It gathers the data concerning the amount of PM2.5, PM10, NO2, CO, and other harmful pollutants in order for people to know the level

of contamination in their surrounding area. The solution provides sensor-level accuracy and analytical expertise to help organizations and individuals take appropriate environmental action. Clarity Air estimates the pollution level using smart algorithms and offers the forecast regarding the changes in air quality. Being an innovative software platform, Clarity Air can be taken as the reference for commercial air quality monitoring products in terms of performance, accuracy, and display. The comparison table below shows the peculiarities of monitoring air quality products offered by these companies.: undefined [6].

The main features of the Application are listed below:

- Real-Time Monitoring: Provides instant air quality readings using IoT sensors. Pollutant Tracking
- Interactive Air Quality Dashboard: Displays real-time data through engaging visuals, charts, and maps, making it easier for users to understand air pollution levels and track changes effectively.
- Personalized Health Insights: Offers custom health tips and recommendations based on air quality data, ensuring users receive guidance that supports respiratory wellness and safer lifestyle habits.
- Location-Based Air Tracking: Detects users' current location through GPS and delivers precise air quality readings nearby, providing accurate environmental updates wherever they go instantly.
- User-Friendly Customization: Lets users adjust dashboard themes, select pollutants to display, and manage alert settings easily, offering a personalized and visually appealing monitoring experience.
- Smart Integration Support: Works smoothly with IoT air sensors, fitness wearables, and home assistants to provide connected, real-time air quality insights for smarter and healthier living environments.

The limitations of the application are listed below:

- Internet Dependency: Requires constant connectivity. Sensor Coverage
- Data Delay in Low-Monitoring Zones: When operating in areas with limited sensors or poor data transmission, the app might show slightly delayed updates or outdated pollution readings, reducing its real-time accuracy.
- Performance Issues on Older Devices: On older or low-performance smartphones, the app may lag, freeze, or crash, especially when displaying live maps or complex air quality visualizations in real time.
- Premium Features Restriction: Many of Clarity Air's advanced tools—like detailed forecasting, personalized health insights, and long-term analytics—are available only through paid subscriptions, limiting free users' experience.
- Complexity for First-Time Users: The detailed pollutant metrics, interactive charts, and multiple dashboard options can appear overwhelming for beginners unfamiliar with AQI concepts or environmental terms.
- Requires Constant Internet Access: The app depends heavily on an active internet connection to fetch live AQI updates and forecasts, making it less functional or completely inaccessible in offline conditions.

IQAir AirVisual: It is a software that monitors the air quality. It is operated by IQAir AG, an independent company that utilizes both ground and satellite sensors to provide its users with accurate data on air quality. It analyses the levels of pollutants such as PM2. 5, PM10, CO2, and O3. AirVisual is an interactive air pollution map that displays the levels of pollution in cities worldwide. It helps people understand what is happening around them and how they can help reduce pollution. The software uses cloud technology to provide its users with constant data streams about air quality and pollution. IQAir AirVisual helps in understanding commercial air monitoring software in terms of accuracy, user interface, and prediction of pollution levels in the best places around the globe. [7].

The main features of the Application are listed below:

- Real-Time Monitoring: Provides instant air quality updates worldwide. Pollutant Tracking
- Interactive Global Mapping: The system features an interactive map that visualizes pollution levels across multiple cities and countries, allowing users to explore and compare global and local air quality conditions easily.
- Comprehensive Pollutant Measurement: It measures key pollutants such as PM2.5, PM10, CO, and O, enabling users to assess exposure levels and understand potential health risks based on real environmental data.
- Predictive Forecasting and Health Insights: The platform delivers air quality forecasts along with personalized health recommendations, helping users plan outdoor activities safely and minimize exposure to harmful environmental conditions.
- Cloud-Based Data Processing: It utilizes cloud computing to process, store, and update air quality data in real time, ensuring reliability, scalability, and seamless performance across various devices and geographic regions.

The limitations of the application are listed below:

- Limited Sensor Coverage in Remote Areas: Some regions lack sufficient ground-based sensors, causing data gaps or reduced accuracy in monitoring air quality, particularly in rural or less developed geographic locations.
- Dependence on Internet Connectivity: The application relies on stable internet access for real-time updates, which limits functionality in regions with poor connectivity or where network interruptions frequently occur.
- Restricted Access to Historical Data: Users may face limitations in accessing long-term historical pollution records, making it difficult to perform detailed trend analysis or track environmental progress over extended periods.
- Potential Data Inconsistencies: Variations in satellite calibration, local weather conditions, and sensor maintenance can occasionally lead to discrepancies or inaccuracies in reported pollution measurements and air quality indexes.
- Limited Customization for Professional Use: The app focuses primarily on general public awareness, offering fewer advanced data analytics or export options needed by researchers, policymakers, and environmental professionals for detailed assessment.

BreezoMater: Air Quality Monitoring System Essay (Analysis) BreezoMater is an innovative air application created by BreezoMeter Ltd. The product allows users to track pollution levels in their location using satellite images, state pollution monitoring, and artificial intelligence. The algorithm calculates PM2.5, PM10, CO, NO2, and O3 to predict air quality and recommend health protection. It visualizes the results as maps and charts to compare the air quality in two areas. The author emphasizes the importance of such systems due to increased awareness of air pollution's impact on the environment and human health. BreezoMeter's team provides high-accuracy data by combining the information from several sources, using cloud technology, and applying AI algorithms. These algorithms enable the app to analyze the air quality and represent data in a form understandable to non-experts. [8].

The main features of the Application are listed below:

- Real-Time Monitoring: Provides instant air quality updates worldwide. Pollutant Tracking
- Smart Home Connectivity: Connects effortlessly with IoT devices and air purifiers, automating indoor air adjustments based on outdoor pollution levels for improved home air quality.
- AI-Driven Pollution Forecasting: Uses advanced predictive algorithms to estimate future air quality levels, allowing users to plan outdoor activities and reduce exposure to harmful pollutants effectively.
- Detailed Air Quality Maps: Displays interactive, color-coded maps that visualize pollution intensity across regions, helping users easily identify clean and polluted areas in real time.
- Personalized Health Alerts: Sends customized alerts when air quality becomes unsafe, guiding users to take preventive measures such as wearing masks or staying indoors when necessary.
- Weather and Climate Insights: Integrates weather forecasts with air quality data to give users a complete environmental overview, supporting informed lifestyle and travel choices every day.

The limitations of the application are listed below:

- Internet Access: Requires online connection for updated readings. Data Accuracy
- Data Delay in Remote Regions: Air quality readings in less populated or remote areas may experience noticeable delays due to slower sensor updates or limited satellite input, causing temporary mismatches between displayed and actual conditions.
- High Battery and Data Usage: Continuous GPS tracking, live updates, and background data refreshes consume significant battery power and mobile data, which can be inconvenient for users with limited phone resources.
- Limited Sensor Coverage: In regions with fewer monitoring stations or outdated sensors, the app's air quality data may lack precision. This limitation can lead to less accurate readings in rural areas or developing regions where environmental monitoring infrastructure is still emerging.
- Complexity for New Users: While the app offers detailed charts and advanced settings, beginners may find the interface slightly overwhelming at first, especially when interpreting pollutant indicators or customizing dashboards.

- Device Compatibility and Performance Issues: Older smartphones or devices with limited processing power may experience slow loading times, occasional crashes, or lag when displaying high-resolution air quality maps and live animations.

Air Matters: discusses about the effect of indoor air to the people, their comfort, and productivity. It highlights that people can live their lives indoors where there is less ventilation and exposure to paint, cleaning agents, and furniture resulting in poor indoor air quality. It also calls upon building designers and developers to incorporate more effective ventilation systems in the buildings to promote clean indoor air and at the same time free people from exposure to harmful chemicals. The proposal of the app is focusing on the importance of indoor air quality, societies across the globe can ensure clean indoor air which eventually leads to a more productive community and a healthier individual living both at home and in the office. [11].

The main features of the Application are listed below:

- Real-Time Data Display: It displays live readings of temperature, humidity, and pollutant levels on a clear dashboard, allowing users to observe air quality variations instantly and respond before harmful levels increase.
- Air Quality Monitoring: The app monitors indoor air quality by measuring pollutants, dust, and humidity through smart sensors. It continuously tracks changes to ensure users stay informed about their breathing environment.
- Smart Alerts and Notifications: The app sends instant notifications when air conditions become unsafe, warning users to take action. It helps in maintaining safe indoor air by identifying potential risks immediately.
- History and Record Keeping: The app stores historical air data, creates visual graphs, and allows users to compare trends over time, helping them evaluate progress and maintain consistently cleaner air indoors.
- Connectivity and Sharing: It connects to mobile devices via Bluetooth or Wi-Fi and enables users to share detailed air reports, promoting awareness and encouraging healthy indoor habits among others.
- Air Quality Monitoring: The app monitors indoor air quality by measuring pollutants, dust, and humidity through smart sensors. It continuously tracks changes to ensure users stay informed about their breathing environment.

The limitations of the application are listed below:

- Sensor Accuracy Limitation: The app depends on external sensors that sometimes produce slightly inaccurate readings due to temperature shifts or device calibration, which affects the precision of recorded air quality data.
- Connectivity Issues: It relies on stable Bluetooth or Wi-Fi connections. Weak signals or interference cause delayed updates and limit the app's ability to display real-time data consistently.
- Limited Pollutant Detection: The app detects only selected pollutants such as dust or carbon dioxide and misses certain harmful gases or chemicals that require advanced industrial-grade sensors for complete detection accuracy.

- Power Consumption: Continuous monitoring uses significant battery power, especially when sensors run for extended periods, which reduces overall device battery life and requires frequent recharging.
- Environmental Dependence: Performance varies depending on environmental conditions like humidity or temperature, as these factors influence sensor behavior and affect the reliability of air readings in some areas.
- Data Privacy Limitation: The app stores user data for analysis, which raises privacy concerns if not properly secured. Unauthorized access poses risks to personal information and environmental records.

Table 2.1 presents an overview of the Plume Labs: Air Quality App, focusing on aspects such as features, performance, user experience, and limitations. It includes details on air quality data, usability, battery usage, and regional coverage. User ratings and feedback provide a clear comparison between the free and premium versions.

Table 2.1: Applications Comparison

Features	Applications					
	AirGuard [5]	Clarity Air [6]	IQAir AirVisual [7]	Breezo Meter [8]	Air Matters [11]	Proposed System
Real Time Air quality Monitoring	✓	✓	✓	✓	✓	✓
Personalized Health Recommandations	✓	✓	✗	✗	✗	✓
Street Level Pollution Maps	✓	✗	✗	✗	✓	✓
Air Quality Forcasting	✓	✓	✓	✓	✓	✓
Air Pollution Health Impact Information	✓	✓	✗	✗	✗	✓
Global Coverage	✗	✗	✓	✓	✗	✓
Social Sharing	✗	✓	✓	✗	✗	✓
Real Time Alert and Notifications	✓	✗	✗	✗	✓	✓
Indoor Air Quality Monitoring	✓	✗	✗	✗	✓	✓
Historical Data	✗	✗	✗	✓	✓	✓

2.2 Future System Usage Analysis

The system will be used to monitor the state of the art in time and make predictions about the state of the art in different areas. It will help government authorities and health care institutions and urban planners and environmental researchers make relevant choices for the available data. The system will employ solutions such as artificial intelligence, high-speed internet, and specialized computer systems to ensure the information is retrieved faster and the forecasts are more accurate. It will be used in urban areas and small towns and will be of great significance to both locations. It will also leverage applications that are easy to use and will ensure collaboration is achieved among the involved parties in monitoring the state of the art. It will also help planners make the best choices and help government implement the proper regulations regarding the environment. The air quality monitoring system will be of great importance, making the

world a better place. The air quality monitoring system will lead to the development of healthy and intelligent communities that are favorable to the environment. The air quality monitoring system will be of global significance.

2.3 Problem Statement / Limitations

Air pollution is dangerous to human health and the environment. The current approaches are not sufficient in allowing the general public to be aware of what is happening in their environment. It affects the response time to medical aid regarding pollution-related diseases, ineffective city management plans, and poor response to environmental threats. The system also monitor the presence of harmful pollutants in the air using sensors and computer models and give predictions based on its readings. The system helps in giving timely warnings concerning the air quality as well as tracking changes in the air quality over time. The system also allow citizens to be more educated about their environment and have effective decision-making in each location that they live.

The system has several limitations that address planning and developing the proposal. The accuracy of sensors depends on the prevailing weather conditions and the topography of the site where they are installed. Sparsely populated areas have connectivity issues that make it hard to ensure a steady flow of information and timely insights. The use of external sources, including OpenWeather and IQAir, poses challenges in terms of latencies and inconsistencies in data formats that hinder unified forecasting. The computational power needed for AI and machine learning algorithms lead to delays in generating alerts. To resolve these issues.The design allow storing data offline and implement mechanisms for data synchronization. The creation of a uniform framework for interacting with external APIs standardize data collection.

2.4 Proposed Solution

The proposed solution involves the creation of an innovative and intelligent system for monitoring and forecasting air quality levels. The system would leverage technology and artificial intelligence capabilities to effectively tackle the outlined problem. Some of the key aspects of the solution include advanced sensors for identifying air pollution levels, as well as big data analytics tools capable of analyzing and interpreting the results. These include policymakers, researchers, government officials, and the general community. The solution offers an intuitive view on the level of pollution in different parts of the country, as well as the forecast for the coming days, based on the analysis of the sensors. The proposed system also considers other environmental factors that may contribute to air pollution, such as weather system, temperature, wind patterns among others. Integrated system using application programming interfaces from other reliable sources, including open-weather and IQAir. The system has an advanced system that can detect and track pollutants, including PM2.5, PM10, CO2, amongst others. It is through these feature that the system can provide health advisory services, as it can predict the effect of the pollution levels on individuals' health. The proposed solution addresses the main problem by ensuring that people are aware of the pollution levels in their respective environment, thus bringing about the necessary adaptations to promote healthy living.

2.5 Software Process Model

A software process model is a process that describes the various steps, activities, and procedures used to develop a software system. The process model ensures quality output and standardized results by maintaining order and control over the various aspects of the development process. A software process model standardizes requirement definitions, design, implementation, testing, and maintenance activities. Using a process model facilitates the ability to accomplish tasks and activities needed to develop software systems while meeting the set standards and delivering products that satisfy users requirements and expectations. [9].

The selected software process model is the Agile model. The rationale behind the choice of the particular model is its emphasis on flexibility and constant development of the software in question. It also helps achieve a higher level of responsiveness.

2.5.1 Introduction

The Agile methodology in software development is used because of its iterative and incremental approach which focuses on adaptation and improvement. It breaks down development into smaller segments known as sprints. The sprints promote rapid development and improvements. The Agile methodology is effective for software development because of constant customer feedback and updates. The model ensures that developments are user-oriented since constant feedback is utilized to improve the product. It also ensures that testers and developers are working together effectively. Agile is flexible because it allows for adaptations to take place which enables developers to modify requirements. The Agile approach is most appropriate for software development because it allows various methodologies such as Scrum and Kanban to be incorporated. For example, the proposal Scrum works by ensuring that developers create products in each sprint whereas Kanban helps to visualize the work process thus promoting effective task management. Agile also effective for software development as it ensures that the process is efficient while also minimizing risks. The model ensures constant testing and integration of features which leads to quality software development. [10].

Figure 2.1 Illustrate the workflow of the development process, showcasing key stages such as analysis, design, implementation, testing and deployment. It visually represents how phrase interconnected.

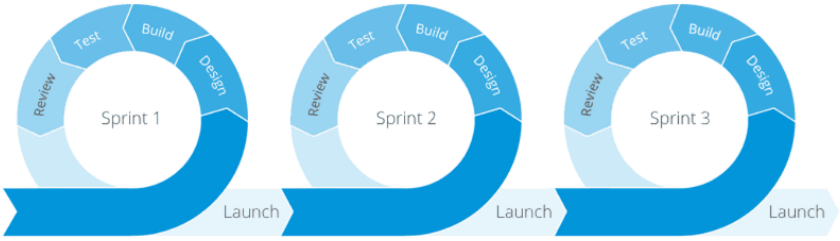


Figure 2.1: Agile Sprint Model [10]

2.5.2 Justification

Agile will be selected because of its flexibility and responsiveness to changing needs. It will lead to incremental changes that meet the end-users' expectations while managing the proposal risks. The approach will also result in continuous improvement as a means of resolving potential problems before they will become serious, thereby producing high-quality software. Agile will increase the stakeholders engagement throughout the process, resulting in fewer conflicts and miscommunication. Most importantly, Agile will lead to software that will be enhanced consistently to fulfill users' needs throughout the process. Since it will allow for the creation of a product that works and meets the users' expectations, it will be selected over linear models. Agile will also appropriate for all software sizes and complexities, thus being the preferred software development methodology. Agile will enhance the processes to help the team in questionhave some of the most flexible and user-ready software while also delivering the best value required by stakeholder in the shortest time. Another advantage is that it will promote transparency and visibility, thereby helping the software team adapt to the changes as they occur.

2.5.3 Steps

The development approach is based on the Agile software model which focuses on iterative development and constant feedback within the process. Which allows for greater flexibility and efficiency within the work while also promoting high-quality results which accommodate the end-user needs.

Requirement Gathering and Analysis: The project team will identify the key functional and non-functional requirements by conducting stakeholder analysis to understand their needs and expectations. The project will focus on features such as pollutants, health alerts, and forecasting. Sprint Planning

Chapter 3

Software Requirements Specification

A Software Requirements Specification (SRS) is a detailed document that defines the functional and non-functional requirements of a software system. It serves as a blueprint that guides developers, designers, and stakeholders throughout the development process. The SRS outlines what the system must do, how it should perform, and the constraints it must operate within. It ensures clarity, minimizes misunderstandings, and provides a foundation for design, implementation, and testing. By maintaining consistency and alignment, the SRS plays a crucial role in delivering reliable, high-quality, and user-focused software solutions [12].

The following sections cover functional requirements, non-functional requirements, user interface design, system features, and performance metrics of the air quality monitoring system. These aspects align with the system's objectives and serve as a foundation for system design and development. Each section defines the expected behavior, usability considerations, system constraints, and technical specifications required to deliver a reliable, accurate, and user-friendly solution.

3.1 Introduction

The SRS document will present the basis for the real-time air quality monitoring system. It will describe technical requirements for developing the product, guide the process to meet all major objectives, and support the design, testing, and evaluation of the future system. As a result, an efficient system for monitoring air quality will be designed.

3.1.1 Document Scope

The SRS document describes the scope of the real-time monitoring system for air quality by indicating the system's features, interfaces, data processing methods, and integration capabilities. The document also specifies its reliability, performance, and security aspects. The SRS document excludes the details related to the manufacturing process of the equipment and the information systems owned by other organizations.

3.1.2 Audience

The SRS document is intended for developers, architects, testers, and stakeholders involved in the air quality monitoring system. The given paper is supposed to help them identify all requirements and keep the common vision of the proposal up-to-date. Below are the main stakeholders' benefits from the detailed description provided in the document.

- **Developers:** Will use to know how it should work accurately to build the system correctly and efficiently. System Architects

3.2 Functional Requirements

Functional Requirements (FR) define the specific actions, behaviors, and operations a system performs to meet user and organizational needs. They describe how the system responds to inputs, processes data, and delivers outputs, ensuring each feature serves its intended function. FR play a vital role in software development by providing a structured understanding of what the system must accomplish. In the Air Quality Monitoring and Prediction System, these requirements guide user interaction with the web platform and determine how sensor data is gathered, analyzed, and presented effectively. [13].

Table 3.1 shows the software's functional requirements. Each module and it's feature must meet specified functional criteria in order for the proposed system to perform smoothly and efficiently Table 3.1 undefined

Table 3.1: Functional Requirements

Requirement ID	Description
FR1	User can create an account on the system.
FR2	User can login to the system.
FR3	User can view personal profile information.
FR4	User can update profile information.
FR5	System collects real-time air quality data from connected sensors.
FR6	System processes sensor data securely in a database.
FR7	User can view current air quality status on a dashboard.
FR8	User can view historical air quality data and trends.
FR9	System predicts future air quality using collected data and prediction algorithms.
FR10	User can view alerts for specific air quality thresholds.
FR11	System can notify users via email or in-app notifications when thresholds are exceeded.
FR12	User can export air quality data in CSV or PDF format.
FR13	System provides a visual representation of data (charts, graphs, maps).
FR14	User can give feedback on system performance or report sensor issues.
FR15	User can logout from the system.

3.3 Non-Functional Requirements

Non-Functional Requirements (NFR) define the quality attributes and performance standards that determine how a system operates rather than what it does. They describe factors such as reliability, security, usability, efficiency, and scalability, ensuring the system functions effectively under various conditions. NFR play a crucial role in software development by establishing benchmarks for system quality and user satisfaction. In the Air Quality Monitoring and Prediction System, these requirements ensure the platform runs smoothly, processes data accurately, protects user information, and maintains consistent performance during continuous monitoring operations. [14].

The following Non-Functional Requirements sections covers the main aspects of the proposal including performance, reliability, usability, security, scalability, and maintainability. The requirements below relate to the Air Quality Monitoring and Predicting System, which ensures sufficient performance of data processing and other functions, being reliable and user-friendly, as well as addressing security and other aspects of high-level performance.

3.3.1 Software Quality Attributes

Software quality attributes are what make a software product successful in terms of its functioning and what benefits it brings to its users. These attributes help in ensuring that the Air Quality Monitoring and Prediction System is efficient, user-friendly, scalable, reliable, and secure during its functioning and processing of data.

- **Performance:** Ensures responsive and timely data handling, providing up-to-date air quality information. Processes sensor inputs efficiently to maintain smooth system operation.
- **Reliability:** Supports continuous operation without interruption or system failure. Maintains consistent functionality under normal and peak conditions.
- **Usability:** Offers an intuitive and accessible user interface for all users. Simplifies navigation, data interpretation, and interaction with system features.
- **Security:** Protects user data and ensures system integrity at all times. Implements safeguards against unauthorized access and potential breaches.
- **Scalability:** Accommodates additional sensors or users without performance degradation. Adapts system resources to meet increasing demand seamlessly.
- **Maintainability:** Facilitates updates, debugging, and system improvements efficiently. Ensures ease of future enhancements and resolves problems effectively.

3.3.2 Performance Requirements

The software should be capable of processing real-time sensor data with minimal latency, ensuring timely updates of air quality readings and predictions. It should be able to handle simultaneous data streams from multiple sensors without degradation in response time. The system should support efficient storage and retrieval of historical data for analysis and trend reporting. It should be optimized to perform complex computations, such as pollution forecasting algorithms, within acceptable timeframes. Additionally, the software should be able to scale resource usage dynamically to maintain consistent performance under varying loads.

3.3.3 Safety Requirements

The software should be designed to minimize risks associated with incorrect air quality readings or system malfunctions. It should be able to detect sensor anomalies or data inconsistencies and provide alerts to users or administrators promptly. The system should be capable of handling unexpected errors gracefully, preventing data loss or corruption. It should be resilient against software crashes, ensuring recovery without compromising stored information. Additionally, the software should be able to log critical events and system errors for analysis, supporting proactive maintenance and timely corrective actions. These measures should ensure safe and reliable operation for all users.

3.3.4 Other Non-Functional Requirements

The software should be compatible with multiple web browsers and devices, ensuring accessibility for all users. It should be configurable, allowing administrators to adjust system settings, thresholds, and reporting intervals as needed. The system should be interoperable with external databases and APIs to support data integration and analytics.

It should be easily deployable and installable, minimizing setup complexity. Additionally, the software should provide clear documentation and help resources for users and developers. It should be flexible to support future feature expansions without major redesigns, maintaining long-term adaptability.

3.4 Requirements Gathering Techniques Used

Requirements gathering techniques are systematic methods used to identify, collect, and document the functional and non-functional needs of a software system. They play a critical role in shaping the final product by ensuring developers, stakeholders, and users share a clear understanding of expectations. These techniques help uncover user preferences, system constraints, and performance goals, guiding design and development decisions. For the air quality monitoring system, they ensure accurate, reliable, and user-focused solutions. Common techniques include interviews, surveys, observations, workshops, and prototyping, which enhance clarity, reduce misunderstandings, and improve success. [15].

The development of the proposed system involved the use of various requirements gathering techniques to identify and document the requirements. The techniques were discussed in the following section including interviews, surveys, observations, workshops, and prototyping. The discussion highlighted the purpose of each method in gathering the requirements and addressing the scope and expectations of the proposed system.

3.5 Time Frame

A time frame is a crucial component that defines the period during which various phases and activities are expected to occur. In the application, time frames are not merely abstract concepts but foundational elements that guide the planning and execution of each platform aspect. Establishing well-defined time frames serves as a cornerstone for effective platform management. Clear time frames enable judicious allocation of resources, systematic progress monitoring, and the setting of realistic expectations, ensuring structured and efficient execution throughout the platform's development. [16].

Table 3.2 Specifies a detailed time frame for creating the proposed system, outlining each phase of development and ensuring that all tasks are completed in accordance with the planned timetable Table 3.2 undefined

Table 3.2: Time Frame

Phase	**Duration**
User Login and Management	30 days
Sensor Data Storage and Real-Time Air Quality Data	30 days
Live Dashboard	30 days
Data Processing and Cleanup	30 days
Live Sensor Location Tracking and Alerts and Notifications	30 days
Report Generation, AI/ML Smog Prediction	30 days
Health	Safety Insights
30 days	
User Feedback	Complaints, Offline Mode

Continued on next page

Table 3.2 – *Continued from previous page*

Phase	**Duration**
Dark Mode	30 days

Chapter 4

Software Design Specification

The software design specification ensures the creation of an efficient and modular system for monitoring the air quality. The system utilize IoT sensors for the data on air quality, which process and analyze to provide useful information. The non-functional requirements is high-performance system with fast processing speed, scalable system architecture, and high security to prevent unauthorized access. The system ensures data integrity by the data using encryption tools. Non-functional requirement is the use of optimize data structures such as time-series databases to store the data collect from the sensors and the process data. The system ensure it is reliable with minimal downtime to ensure it is always available for use. The system meet the performance requirements, have security features, and a user-friendly interface. The design specifications enable the production of an efficient system that is flexible to meet the technological needs. [17].

Software design specification describes the general idea of the proposed solution for air quality monitoring and prediction. The description covers topics such as sensor installation and utilization, data collection, analysis, and forecasting using artificial intelligence. Particular attention is paid to the structure and data flow within the system, which provides a better understanding of the work of the software being developed. Various specifications, including performance, data processing, location, power, security, and reliability, are highlighted. The design specification also covers the interface design, describing what reports and notifications look like and what the user interface should promise. The design specifications helps to get an overview of the planned system and prepare for work on technical specifications.

4.1 Entity-Relationship Diagram

An Entity-Relationship Diagram (ERD) demonstrates the data architecture of the proposed database, which is related to the monitoring and predicting of the air quality in the area. The diagram indicates the main elements that is in the database: Entity, Attributes, and Relationships. The ERD serves to understand the database structure and its main components; in the way, it helps different stakeholders and developers to operate the proposal information correctly. There are six main entities: Sensor, Location, Air Quality Reading, Prediction, Alert, and User. The elements imply that in the proposed database, particular attention paid to the attributes of sensors, locations, and air quality readings, as well as the connections between readings and predictions, readings and users, predictions and users, and users and alerts. Sensors register air pollution, including PM2.5, PM10, CO2, NO2, and O3, and with exact locations, Air Quality Reading store the data and date. In turn, the entity of Prediction generates based on the aforementioned readings, Alerts are triggered by particular data with consideration given to users' preferences. Users visualize air quality information, receive alerts, and view predictions. With the help of the provided ERD, one can see the information flow from Sensors to Air Quality Reading and Prediction or Alerts and Users. [18].

[Figure 4.1] Entity-Relationship Diagram (ERD) for the air quality monitoring and prediction system, showing key entities Sensor, Location, Air Quality Reading, Prediction, Alert, and User and their relationships, clarifying data flow, interactions, and database structure for efficient storage and retrieval. Figure 4.1 undefined

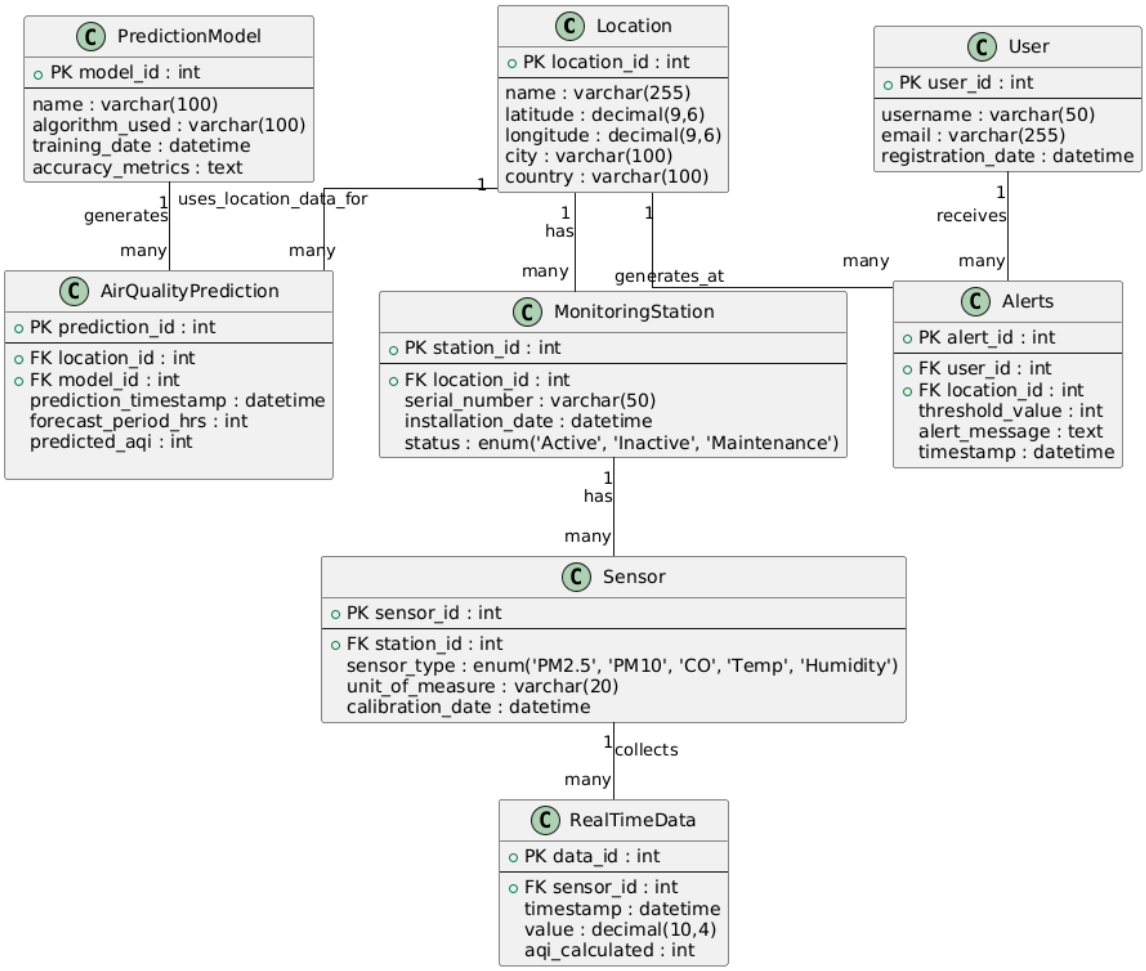


Figure 4.1: ER Diagram of the Proposed System

4.2 Use-Case Diagram

A Use-Case Diagram, in turn, depicts relationships between actors and the air quality monitoring and predicting system under development. It integrates use cases that describe how actors interact with the system. The use cases specify system functions, including receiving air quality data, providing air quality information, notifying of dangerous levels of pollution, calculating air quality forecasts, and analyzing air quality. It focuses on the processes that the system should perform rather than its internal workings. It clarifies the system's boundaries and provides relevant information about its functions and operations. The Use-Case Diagram facilitates collaboration and communication within the development team and with other stakeholders. Users of an air quality monitoring and prediction system include general users and administrators, while external systems are air quality sensors. The diagram specifies the responsibilities and privileges of each actor category. It highlights system features, such as displaying the quality of air, analyzing and calculating air quality, notifying users of dangerous levels of pollution,

receiving and processing data, and providing forecasts. It helps find any missing requirements or functions, as well as unnecessary features. The Use-Case diagram also promotes the creation of documentation, such as sequence diagrams, detailed descriptions of system functions, and other specifications, which are essential for developing the system. [19].

[Figure 4.2] The use-case diagram for the proposed system is the one depicting the behavior of each use case, the interaction between the components, and their roles. The UML diagram assists in understanding how the system works by displaying the connections and interactions between its various aspects and elements, which are critical for proper functioning. Figure 4.2 undefined

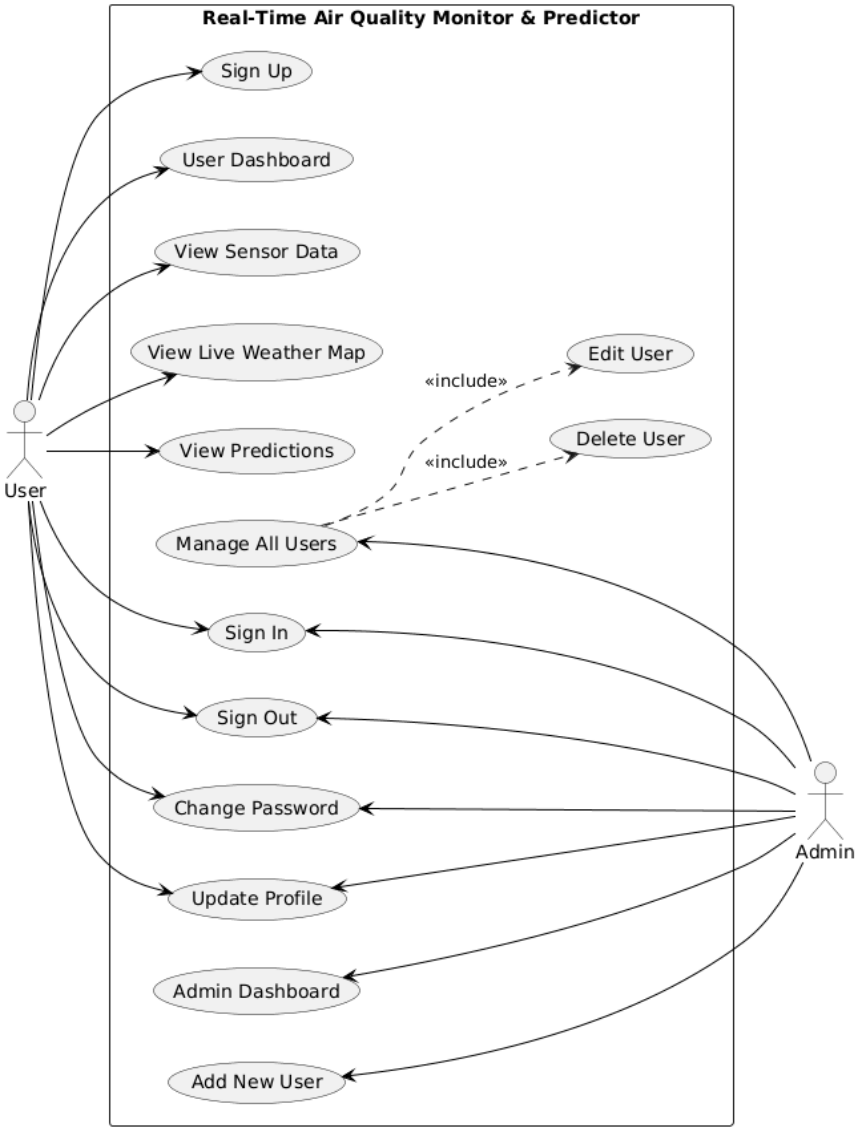


Figure 4.2: Use-Case Diagram of the Proposed System

4.3 Use-Case Descriptions

Use-Case Descriptions offer substantial details regarding the interaction between the actors and the system. Specify the different steps of each identified use case, including the preconditions, postconditions, and alternative and exception paths. The highlight the particular flow of events and operations performed as the system, making it easy

to comprehend how the described functions work. In particular, use-case descriptions explain how the air quality monitoring and prediction system under consideration operate. For instance, they demonstrate how end-users can check air quality, get notified of its changes, review the produced forecasts, and customize their preferences. They offer an extended list of all the necessary activities fulfill in order to develop the system under consideration. As such, use as a reference guide that highlights built of each identified use case. [20].

4.3.1 UC01: SignUp

Actors: User.

Type: Primary.

Brief Description: The use case outlines how a new user registers an account on the platform. It collects essential details such as name, email, and password, verifies them for uniqueness, and securely stores user information for future login and authentication within the system.

Basic Flow:

1. The user visits the registration page from the main interface to begin creating an account. The system loads the form fields that require personal details such as name, email, and password to uniquely identify the user and enable future authentication in a secure way.

Alternative Flow:

1. If the user provides incomplete or invalid information such as a missing email or weak password, the system displays appropriate error messages, prompting the user to correct the mistakes before reattempting submission to ensure proper data validation and consistent registration standards.

Pre-condition:

1. The person attempting registration should not already have an account with the same email or credentials.

Post-condition:

1. A new user record is saved in the system, enabling future login and platform usage.

Exceptional Scenarios:

1. Database connectivity fails during form submission, leading to registration failure.

4.3.2 UC02: Sign In

Actors: User, Admin.

Type: Primary.

Brief Description: The use case defines the process through which a user or an admin accesses their account on the platform by providing valid login credentials. The system ensures that only authorized individuals are granted access to their respective dashboards.

Basic Flow:

1. The user opens the application and selects the "Sign In" option from the main screen. The system displays input fields for username or email and password. Once the user enters credentials, the system immediately validates the provided details. It checks whether the email or username exists.

Alternative Flow:

1. If the user enters an incorrect email or password, the system displays an error message prompting for re-entry.

Pre-condition:

1. The user or admin must already be registered and have an active account in the system.

Post-condition:

1. The actor is successfully logged in, and session management is initiated for secure interaction.

Exceptional Scenarios:

1. If the authentication server is down, login attempts cannot be processed.

4.3.3 UC03: User Dashboard

Actors: User.

Type: Primary.

Brief Description: The user dashboard displays air quality data, predictions, and alerts in a clear layout, enabling users to monitor conditions and access features easily.

Basic Flow:

1. The user logs in, views air quality data, checks predictions, receives alerts, and navigates available features.

Alternative Flow:

1. The user logs in, encounters incomplete data, refreshes the dashboard, adjusts filters, and reviews updated air quality information and alerts.

Pre-condition:

1. The user holds a valid account, completes authentication successfully, and the system loads required data services and permissions properly.

Post-condition:

1. The dashboard displays data, records actions, and allows navigation.

Exceptional Scenarios:

1. None

4.3.4 UC04: View Sensor Data

Actors: User.

Type: Primary.

Brief Description: The view sensor data feature displays real-time and historical air quality readings, enabling users to monitor pollutants and analyze trends efficiently.

Basic Flow:

1. The user selects a sensor, views real-time readings, analyzes data, and navigates through historical records.

Alternative Flow:

1. The user selects a sensor, encounters missing data, refreshes the page, and reviews readings successfully.

Pre-condition:

1. The user logs in successfully, selects the sensor module, and ensures sensors transmit data correctly.

Post-condition:

1. The system displays sensor readings, updates logs, and allows the user to analyze or export data.

Exceptional Scenarios:

1. The system encounters sensor failure or connection error, notifies the user, and prompts data refresh.

4.3.5 UC05: View Live Weather Map

Actors: User.

Type: Primary.

Brief Description: The view live weather map feature displays real-time weather conditions, including temperature, humidity, and wind, allowing users to monitor changes continuously.

Basic Flow:

1. The user opens the weather map, selects a region, views real-time conditions, monitors updates, and interacts with map features.

Alternative Flow:

1. The user opens the weather map, encounters loading issues, refreshes the page.

Pre-condition:

1. The user logs in successfully, accesses the weather map module, and ensures the system connects to live weather data sources.

Post-condition:

1. The system displays real-time weather conditions, updates map visuals continuously, records user interactions, and allows navigation.

Exceptional Scenarios:

1. The system loses connection or fails to load map, notifies user.

4.3.6 UC06: View Predictions

Actors: User.

Type: secondary.

Brief Description: The view predictions feature displays air quality data, showing pollutant levels and trends, enabling users to anticipate changes and plan actions.

Basic Flow:

1. The user selects the predictions module, views air quality data, analyzes pollutant trends, monitors updates, and navigates available prediction features.

Alternative Flow:

1. The user selects the predictions module, encounters incomplete or delayed data, refreshes the page.

Pre-condition:

1. The user logs in successfully, accesses the predictions module, and ensures the system retrieves historical and current air quality data accurately.

Post-condition:

1. The system displays accurate air quality forecasts, updates prediction records, allows user analysis, and supports navigation.

Exceptional Scenarios:

1. The system encounters missing or delayed data, notifies the user of errors.

4.3.7 UC07: Update Profile

Actors: Admin.

Type: Primary.

Brief Description: The update profile feature allows users to modify personal information, including name, email, and password.

Basic Flow:

1. The admin accesses the profile section, edits personal information, submits changes, and the system validates and saves.

Alternative Flow:

1. The admin accesses the profile section, enters invalid information, receives error messages, corrects details.

Pre-condition:

1. The admin logs in successfully, accesses the profile section, and ensures account authentication is verified.

Post-condition:

1. The system saves updated user information, validates changes, reflects modifications.

Exceptional Scenarios:

1. None

4.3.8 UC08: Manage Users

Actors: Admin.

Type: Primary.

Brief Description: The manage users feature allows administrators to add, update, and remove user accounts.

Basic Flow:

1. The administrator accesses the user management module, views user accounts, adds or updates information, assigns roles, and saves changes successfully.

Alternative Flow:

1. The administrator accesses the user management module, encounters invalid input or errors, corrects the information.

Pre-condition:

1. The administrator logs in successfully, accesses the user management module, and ensures proper authentication.

Post-condition:

1. The system updates user accounts, saves changes accurately.

Exceptional Scenarios:

1. None.

4.3.9 UC09: Change Password

Actors: User, Admin.

Type: Primary.

Brief Description: The change password feature allows users to update their account credentials securely, ensuring authentication.

Basic Flow:

1. The user accesses the change password section, enters the current and new passwords, submits the request, and the system updates credentials.

Alternative Flow:

1. The user enters an incorrect current password, receives an error message, corrects the entry, and successfully updates the account password.

Pre-condition:

1. The user logs in successfully, accesses the change password section, and ensures account authentication is verified before entering current and new credentials.

Post-condition:

1. The system updates the user's password securely, validates the change, and allows the user to access the account with new credentials.

Exceptional Scenarios:

1. The system detects incorrect current password.

4.3.10 UC10: Sign Out

Actors: Admin, User.

Type: Primary.

Brief Description: The logout functionality allows a user to securely exit the system. It clears the active session, removes temporary data, and redirects the user to the login or home screen.

Basic Flow:

1. The user or admin selects the "Sign out" option from the dashboard menu. This action is intended to end the session.

Alternative Flow:

1. If the session is already inactive, the system skips session clearance.

Pre-condition:

1. The user or admin must be successfully logged into the system with an active session.

Post-condition:

1. The session is terminated, temporary user data is removed, and the actor is redirected to the login screen.

Exceptional Scenarios:

1. If the server fails to respond during signout, the user remains on the dashboard.

4.4 Sequence Diagrams

Sequence diagrams enable to display relationships between parts or objects of the system and describe their interaction. They describe the operation's course in detail, providing an overview of the object's behavior and determining how it interact with others to achieve certain functions. The tool is helpful to evaluate various aspects of a system, including testing, design, and maintenance. The help of such diagrams, it is possible to ensure the proper functioning of the system, meet all requirements, and eliminate possible errors. Sequence diagrams play a vital role in different stages of the proposal, from analysis and design to implementation and maintenance. Testers, get information on which scenarios be tested, and the stakeholders have a detailed description of the system's work. Such characteristics are necessary to ensure the reliable functioning of the system and operate the software within the framework of the specified requirements. [21].

Figure 4.3 outlines how a user is register on the site. It is an informative section. The site also protects an individual's information by cross-verifying and encrypting the password they provide before allowing them to access the site using the email address they provide and the password they create. Figure 4.3 undefined

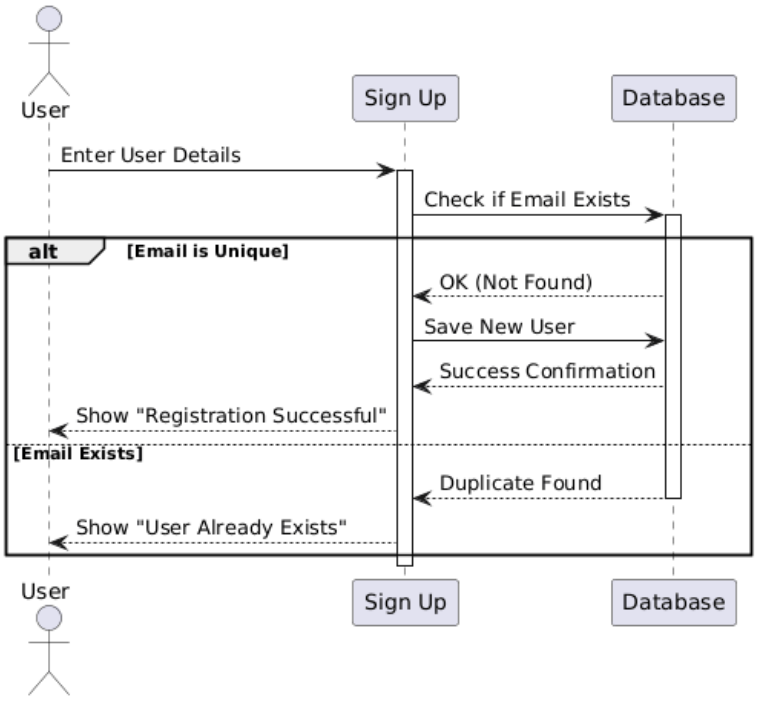


Figure 4.3: Sequence Diagram of Sign Up

Figure 4.4 The user dashboard sequence diagram illustrates interactions between the user, system interface, and database, showing login, data retrieval, display of air quality information, predictions, and alerts in a structured workflow. Figure 4.4 undefined

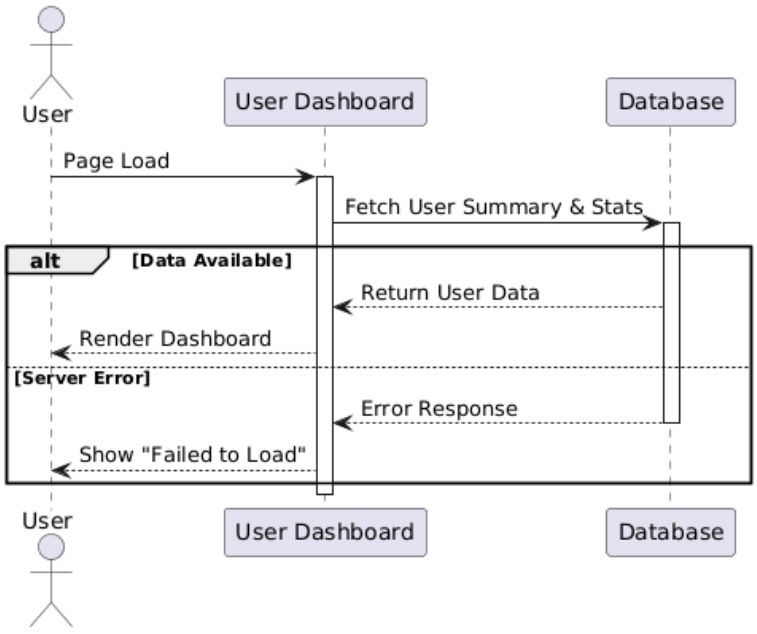


Figure 4.4: Sequence Diagram of User Dashboard

Figure 4.5 The view sensor data sequence diagram illustrates interactions between the user, system interface, and database, showing sensor selection, real-time data retrieval, display of readings, and navigation through historical data efficiently. Figure 4.5 undefined

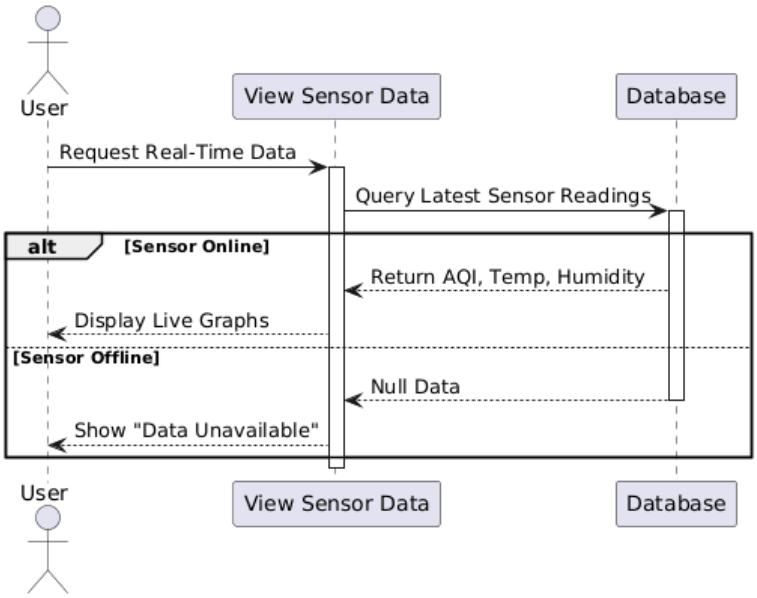


Figure 4.5: Sequence diagram of View Sensor Data

Figure 4.6 The view live weather map sequence diagram illustrates interactions between the user, system interface, and weather data services, showing region selection, real-time data retrieval, map updates, and user navigation through weather information. Figure 4.6 undefined

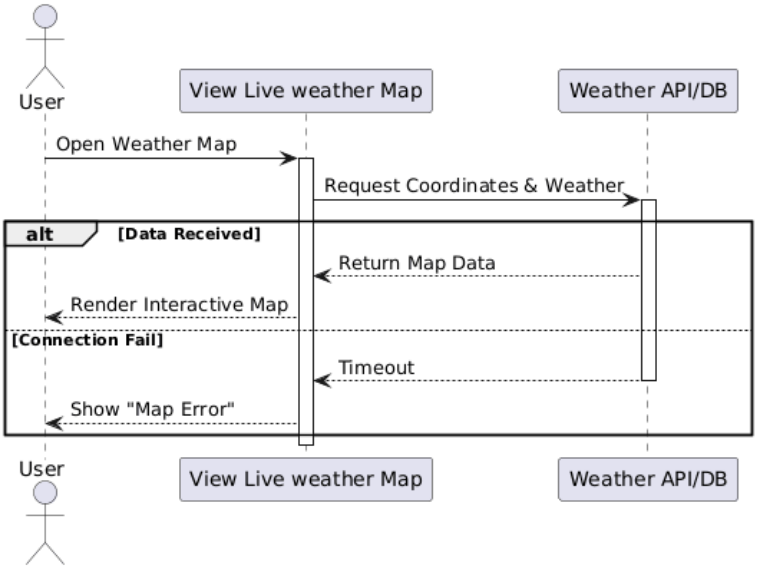


Figure 4.6: Sequence Diagram of View Live Weather Map

Figure 4.7 The view predictions sequence diagram illustrates interactions between the user, system interface, and database, showing selection of the predictions module, retrieval of forecast data, display of air quality trends, and user analysis actions. Figure 4.7 undefined

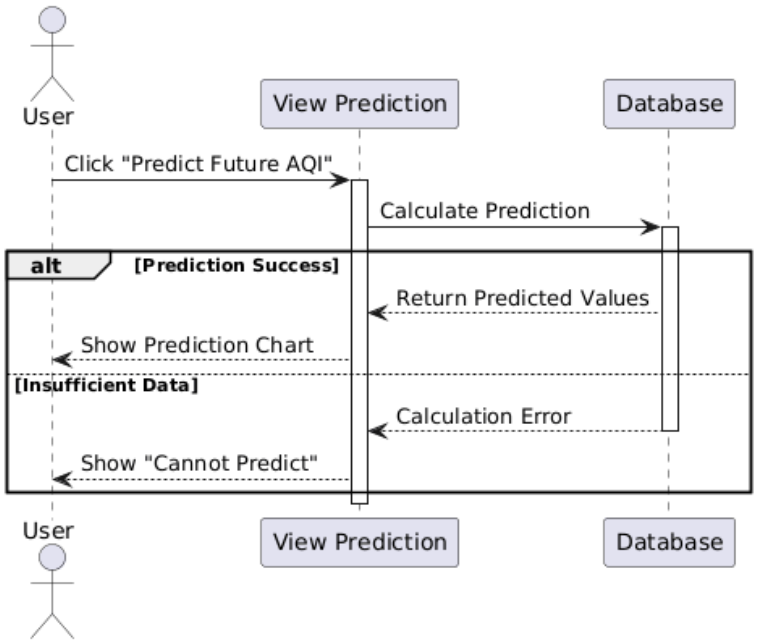


Figure 4.7: Sequence Diagram of View Predication

Figure 4.8 The signin sequence diagram illustrates interactions between the user, system interface, and authentication server, showing user credential input, validation, authentication process, successful login, and access to system features in a structured workflow. Figure 4.8 undefined

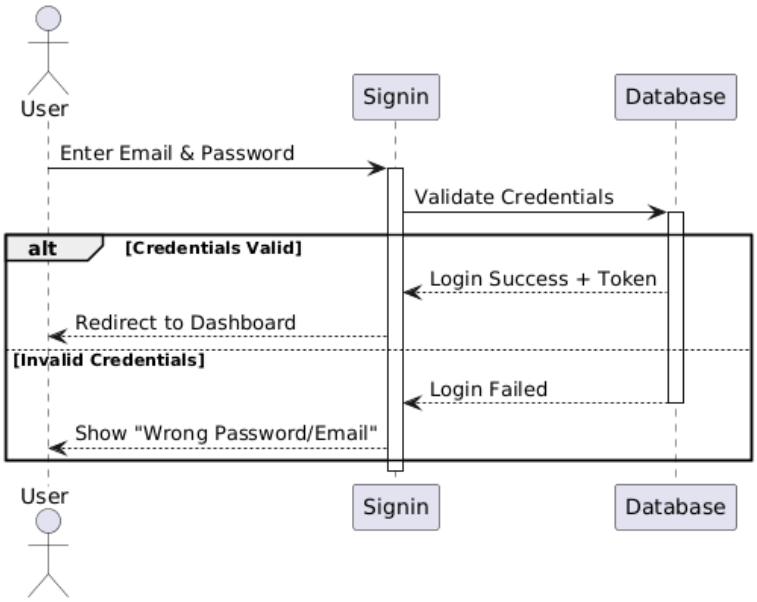


Figure 4.8: Sequence Diagram of signIn

Figure 4.9 The manage users sequence diagram illustrates interactions between the administrator, system interface, and database, showing viewing, adding, updating, or deleting user accounts, assigning roles, and confirming changes in real-time. Figure 4.9 undefined

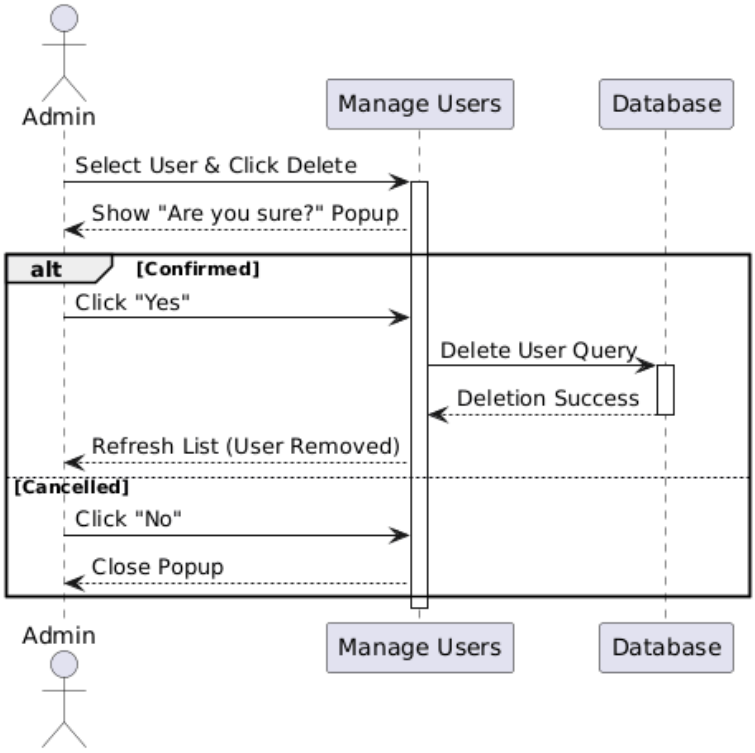


Figure 4.9: Sequence Diagram of Manage Users

Figure 4.10 The admin dashboard sequence diagram illustrates interactions between the administrator, system interface, and database, showing login, user management, data monitoring, notifications, and access to system controls in a structured and efficient workflow. Figure 4.10 undefined

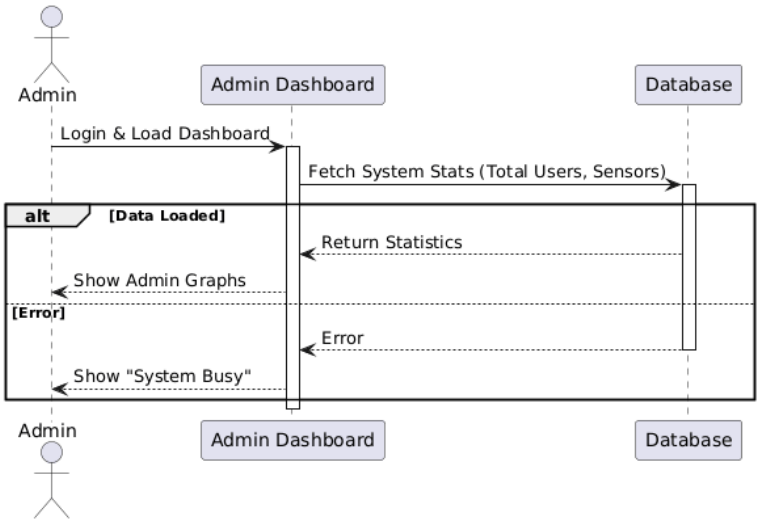


Figure 4.10: Sequence Diagram of Admin Dashboard

Figure 4.11 The update profile sequence diagram illustrates interactions between the user, system interface, and database, showing retrieval of existing information, editing details, submitting changes, validating inputs. Figure 4.11 undefined

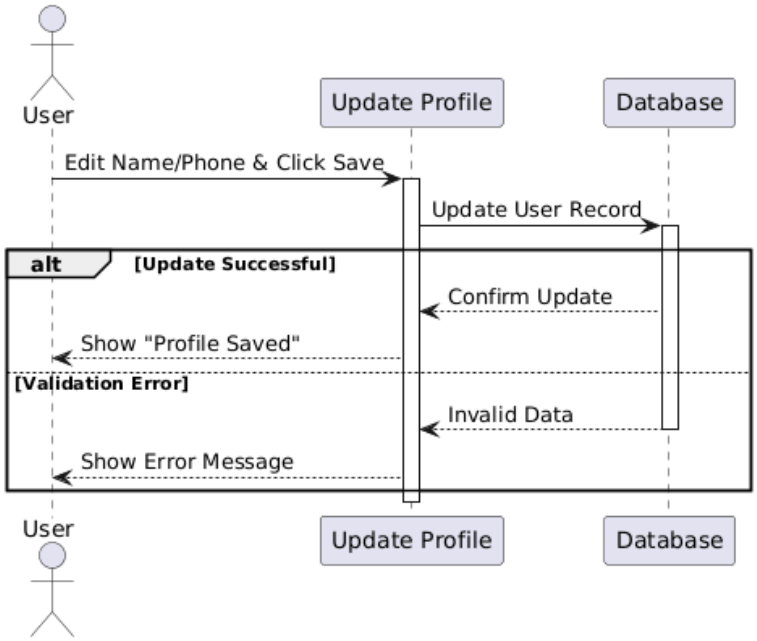


Figure 4.11: Sequence Diagram of Update Profile

Figure 4.12 The change password sequence diagram illustrates interactions between the user, system interface, and authentication server, showing input of current and new passwords, validation, error handling, and successful update of account credentials. Figure 4.12 undefined

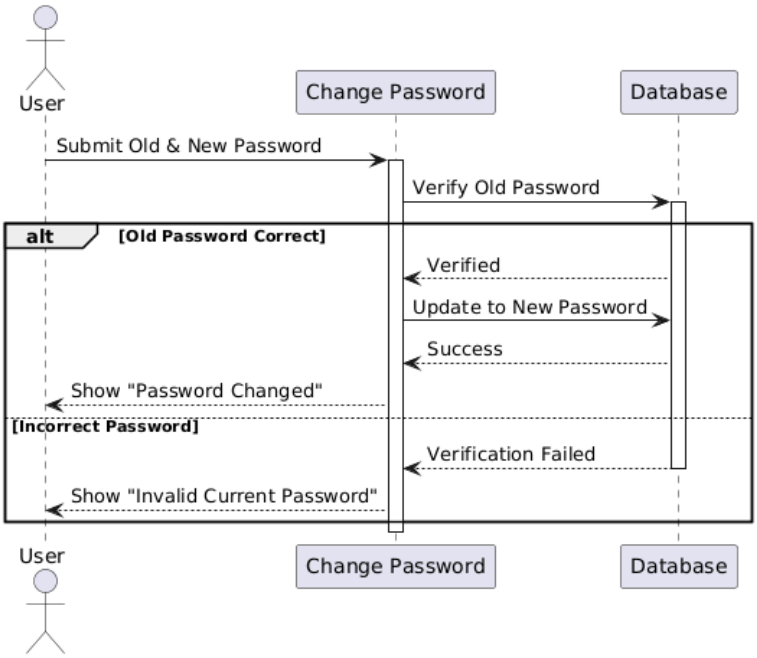


Figure 4.12: Sequence Diagram of Change Password

Figure 4.13 The signout sequence diagram illustrates interactions between the user, system interface, and authentication server, showing the user initiating logout, system terminating the session, clearing credentials, and confirming secure exit from the web application. Figure 4.13 undefined

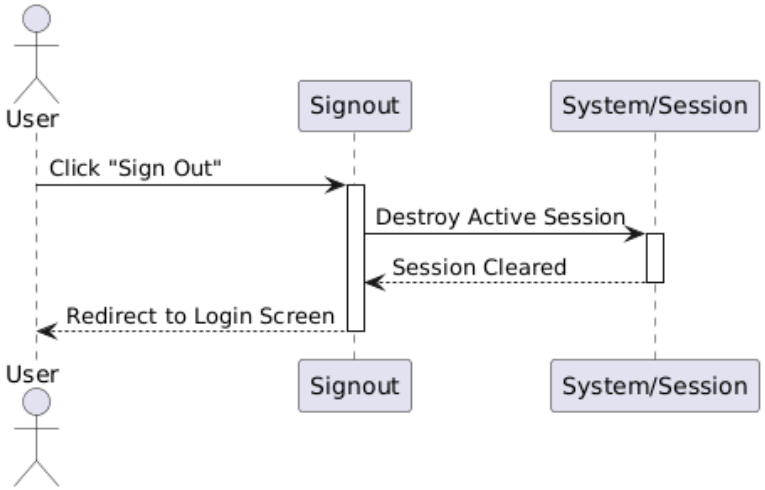


Figure 4.13: Sequence Diagram of Signout

Chapter 5

Interfaces and Physical Design

Regarding the process of developing software, interfaces and physical design refer to the aesthetic and functional components of the user interface, as well as the underlying system architecture. Interfaces describe how users interact with the system, including the layout, controls, and navigation. Physical design entails the layout and integration of system components, databases, and external systems. Interfaces and physical design are critical in the proposed system for producing an intuitive and user-friendly experience. Interfaces that are well-designed allow users to simply move through the program, retrieve essential information, and interact with various features. User engagement and happiness are increased by a visually appealing interface with clear instructions and feedback.Interfaces define how users interact with the system, including elements such as organization, control, and navigation. Physical design focuses on arranging and integrating system components, databases, and external systems. The importance of interfaces and physical design cannot be overstated, as they are essential for providing a clear, efficient, and user-friendly experience. [22].

Interfaces define how users interact with the system, including elements such as organization, control, and navigation. Physical design focuses on arranging and integrating system components, databases, and external systems. The importance of interfaces and physical design cannot be overstated, as they are essential for providing a clear, efficient, and user-friendly experience.

5.1 User Interfaces

The system interface is designed with user-friendly features to support smooth interaction. User interfaces refer to the visual and interactive elements that enable users to communicate with software applications. These elements include panels, menus, buttons, forms, and other graphical components that allow users to input data, navigate through the system, and receive feedback. The interface is user-friendly as it provides a simple and organized platform for interacting with surveying instruments. Such a design leads to superb performance as it allows the user to gain the desired knowledge quickly and master the tool with little effort. The interface offers a helpful tool that enables one to navigate, discover information, and interact with applications easily and efficiently. It is a powerful enabler that facilitates content creation, interaction, and management of tools and their functionalities, providing a dynamic environment where users operate. The interface is paramount in influencing the user's satisfaction with the software web application's performance. Understanding the key characteristics and principles of interface design and being aware of the prevailing industry standards can help in building eye-catching, effective, and interactive interfaces. [23].

The Sign-in screen allows new users to register by entering their email address, and password. It includes input validation and a submit button to create an account. Figure 5.1 undefined

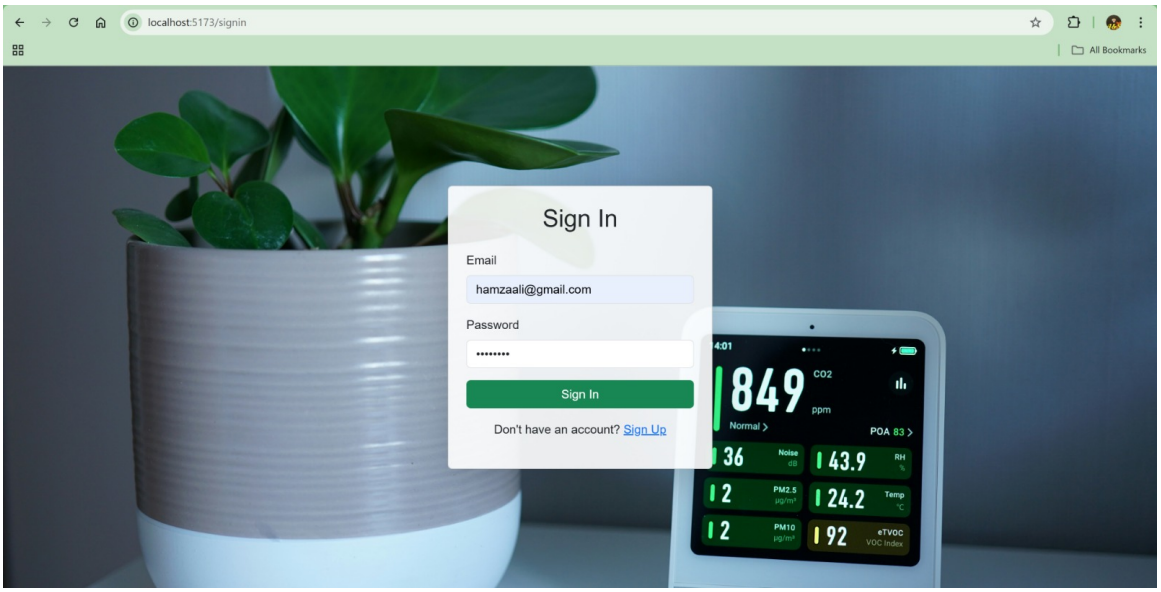


Figure 5.1: User SignIn

The Sign-Up screen allows new users to register by entering their email address, and password. It includes input validation and a submit button to create an account. Figure 5.2 undefined

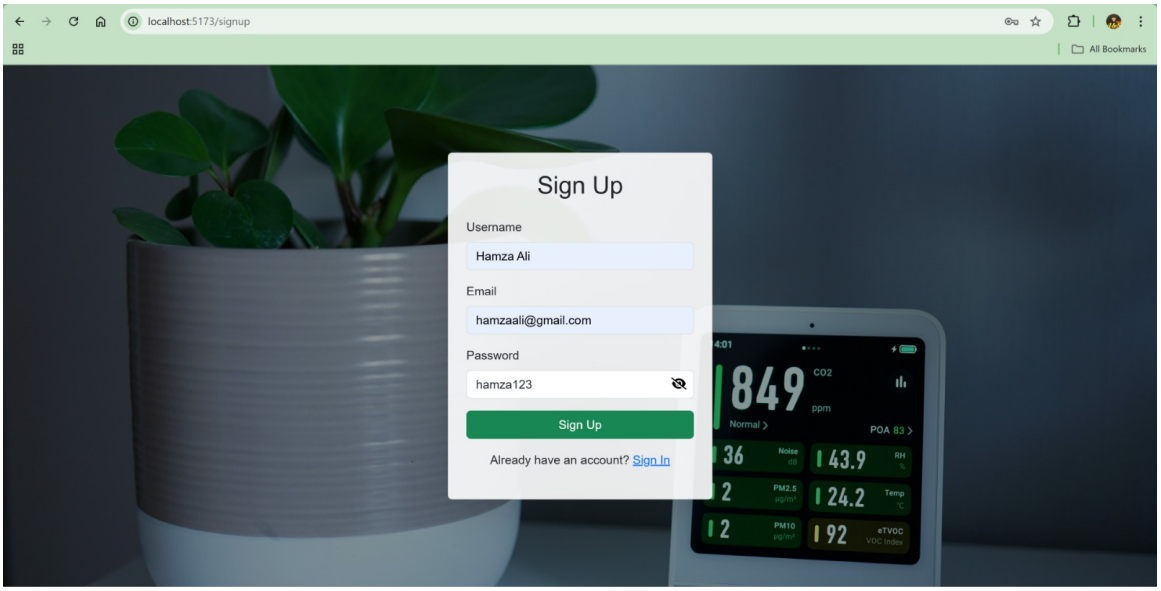


Figure 5.2: User SignUp

The User Dashboard screen allows users to view their screen by entering valid user id. It includes all features control by the user account. Figure 5.3 undefined

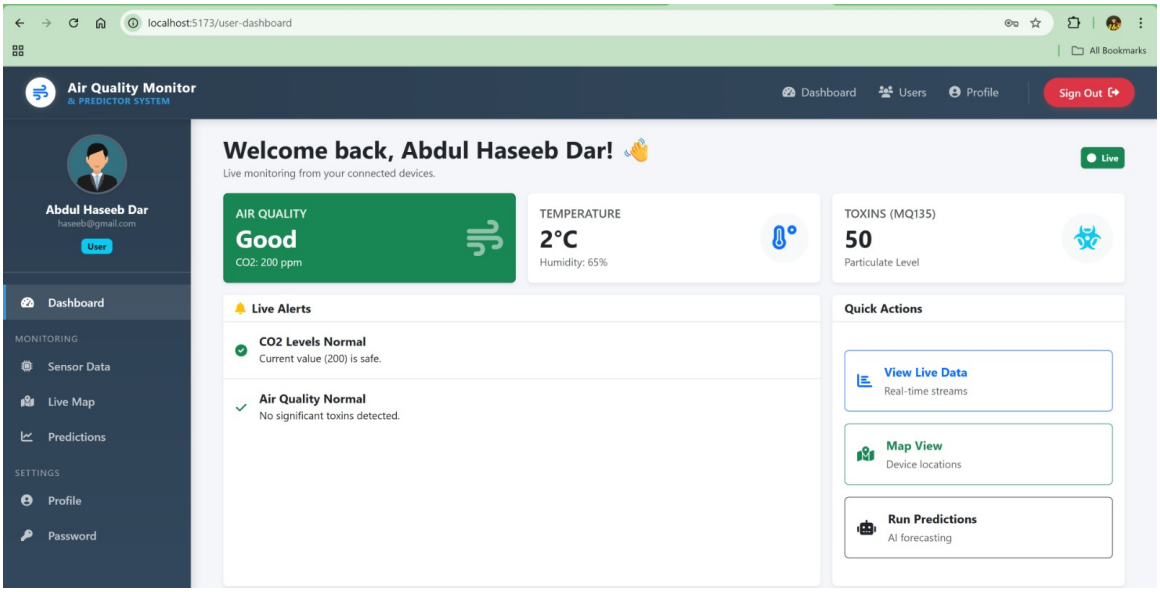


Figure 5.3: User View Dashboard

The User View Sensor data screen allows users to view all sensors connected with the system It includes all sensor data readings user can view. Figure 5.4 undefined

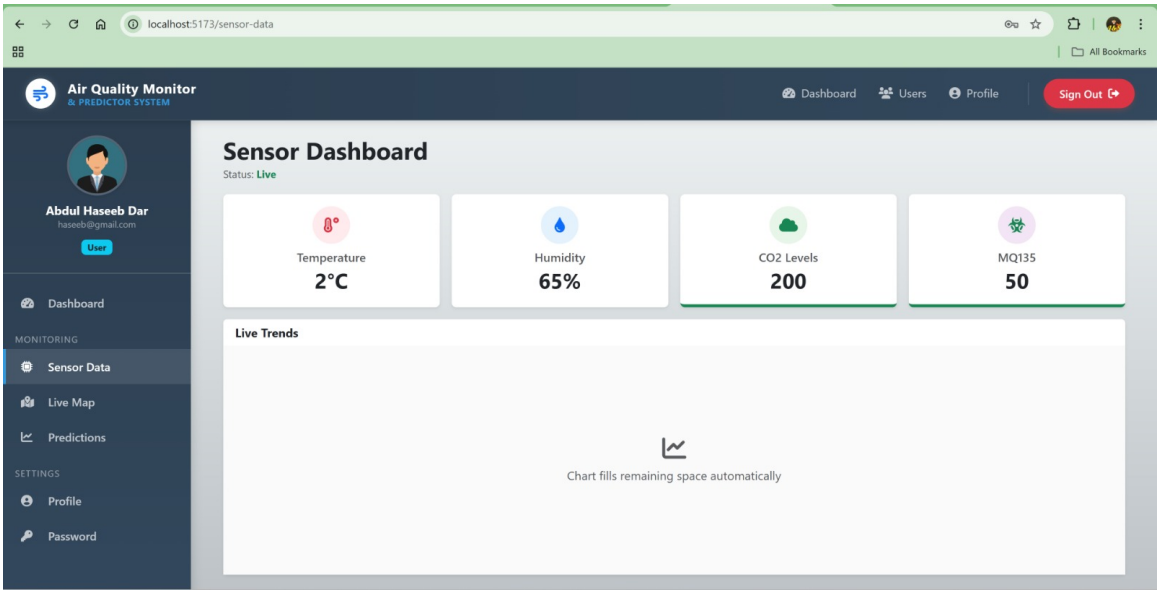


Figure 5.4: User View Sensor Data

The User Live View Weather Data screen allows users to view maps on the screen. It includes global map to see air quality data readings user can view. Figure 5.5 undefined

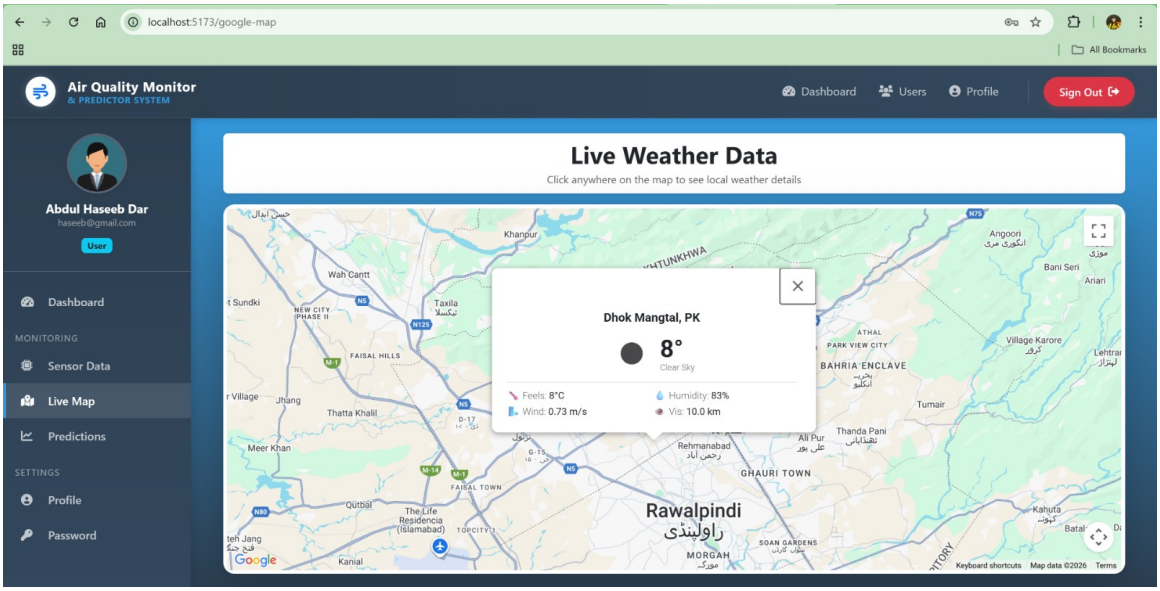


Figure 5.5: Live Weather Data

The User use view predictor screen which allows users to view predications on the screen according to data provided to the sensors in real-time. Figure 5.6 undefined

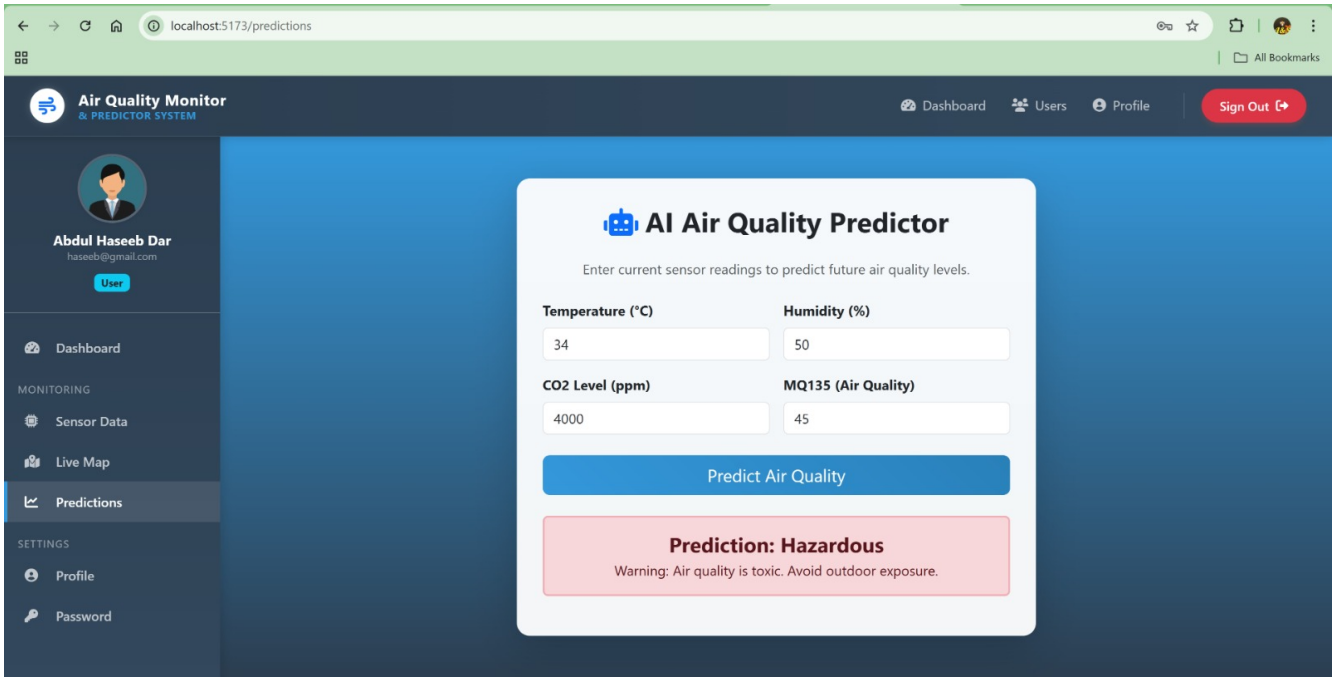


Figure 5.6: View Predications Screen

The User can update profile information in to the system which allows users to view changes on real-time according to data entered in to the system. Figure 5.7 undefined

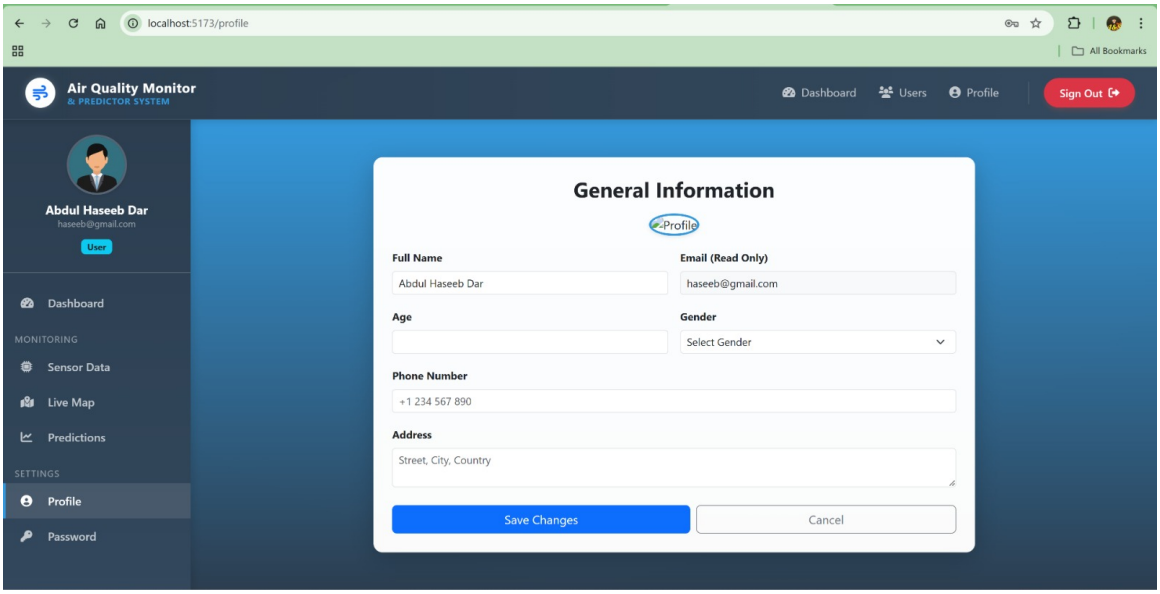


Figure 5.7: User Update Profile Screen

The User can update password in to the system which allows users to view changes on real-time according to data entered in to the system. Figure 5.8 undefined

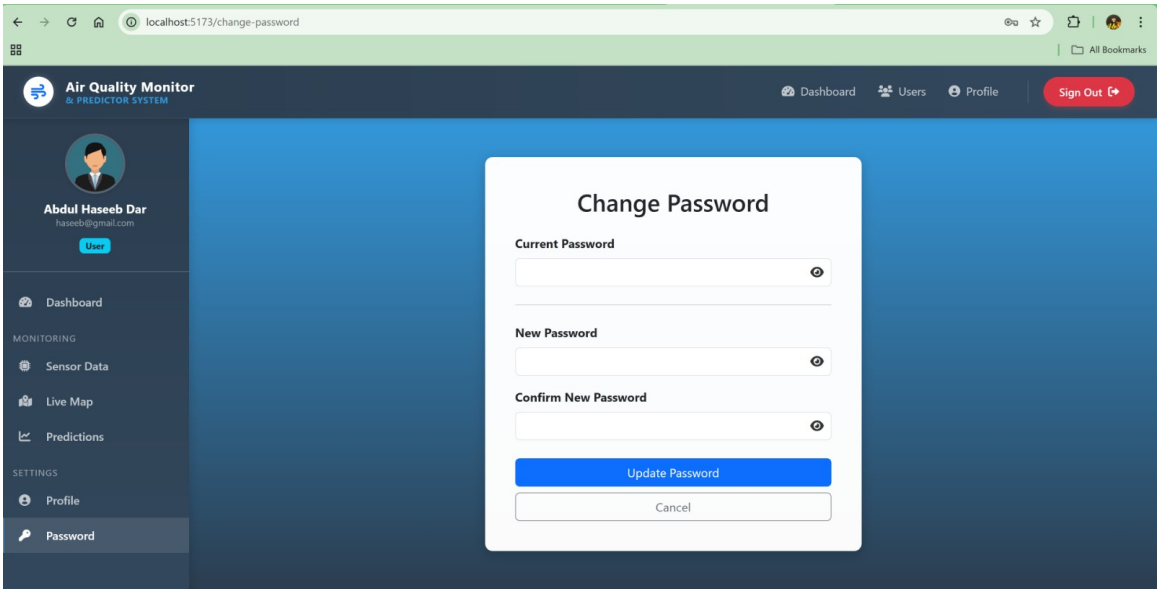


Figure 5.8: User Password Changing Screen

The Admin Dashboard screen allows admin to view their screen by entering valid user id. It includes all features control by the admin user. Figure 5.9 undefined

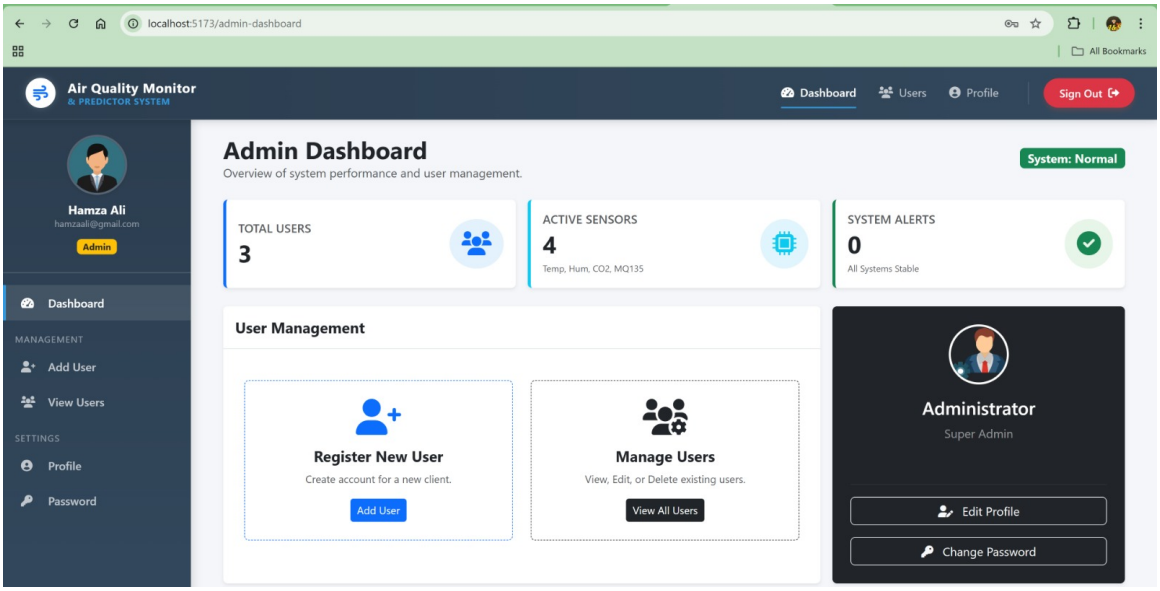


Figure 5.9: Admin Dashboard Screen

The Admin user registration screen allows admin to register any user in the system by entering valid user id. It includes all rights to control by the admin. Figure 5.10 undefined

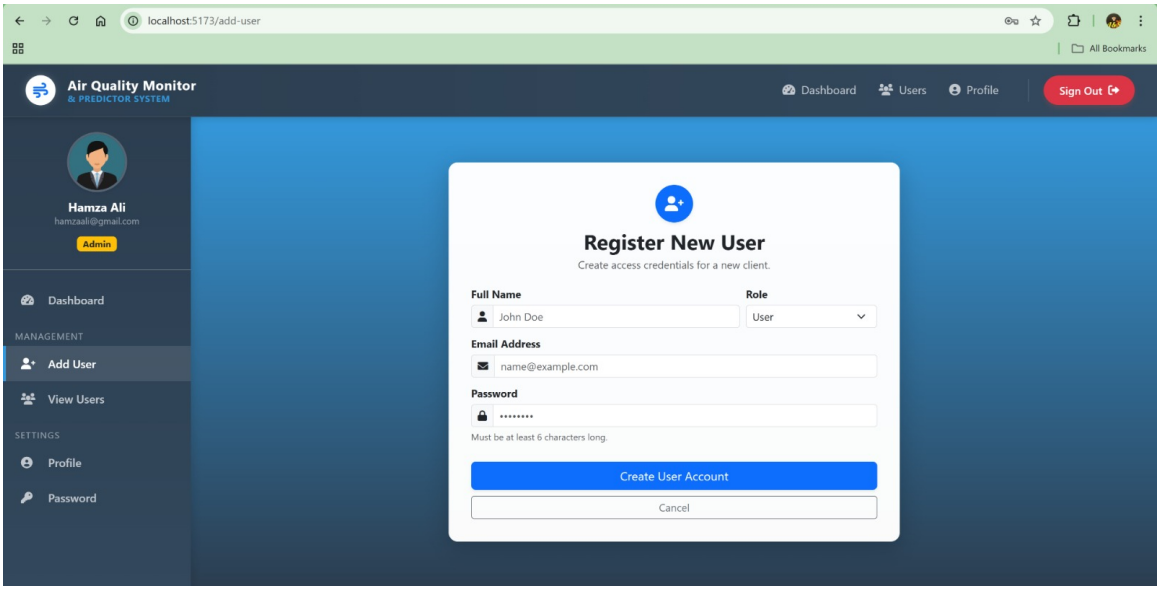


Figure 5.10: Admin Add User Screen

The Admin view or delete user screen which allows admin to view any users registration id and give rights to delete user registration id. Figure 5.11 undefined

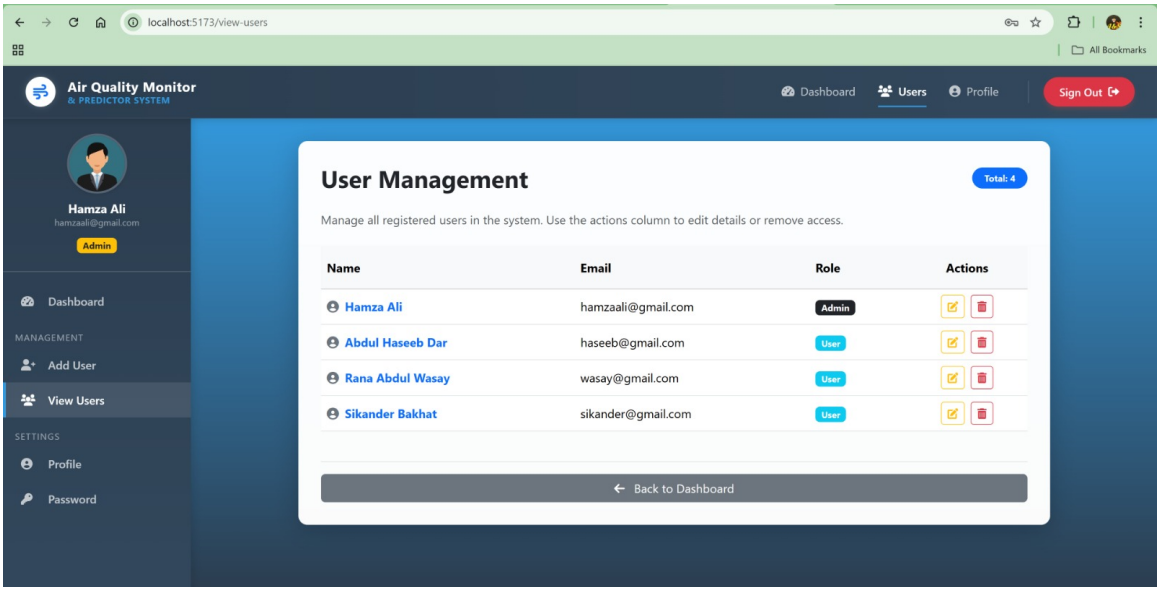


Figure 5.11: Admin View or Delete User Screen

The Admin update profile screen allows admin to update any information of admin in the system by entering valid user id. It includes all rights controled by the admin. Figure 5.12 undefined

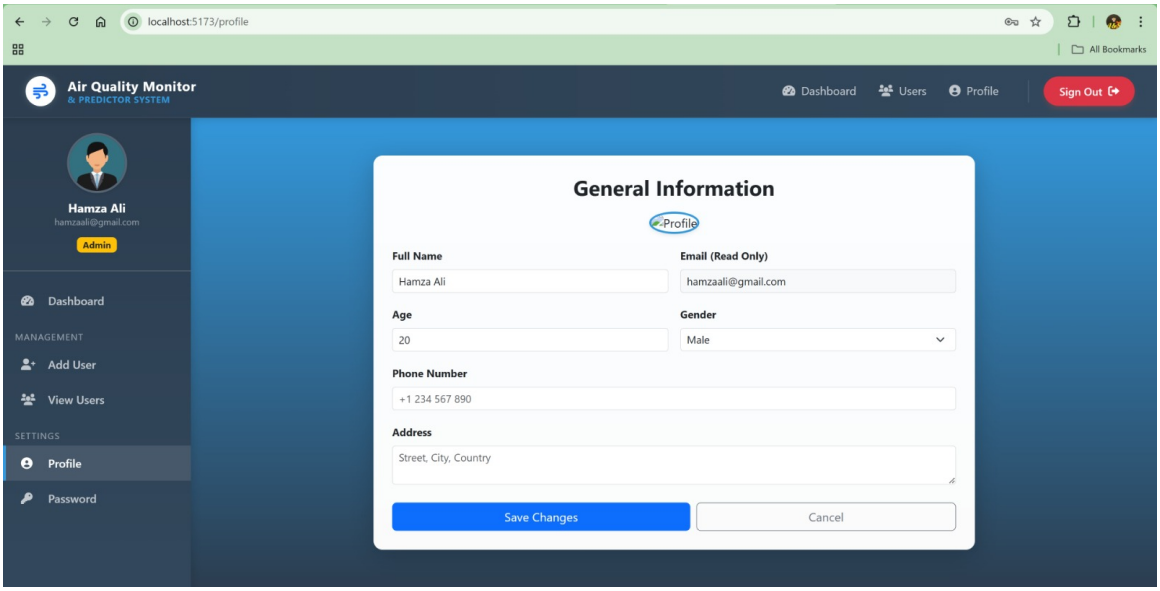


Figure 5.12: admin Update Profile Screen

The admin can update password in to the system which allows admin to view changes on real-time according to data entered into the system. Figure 5.13 undefined

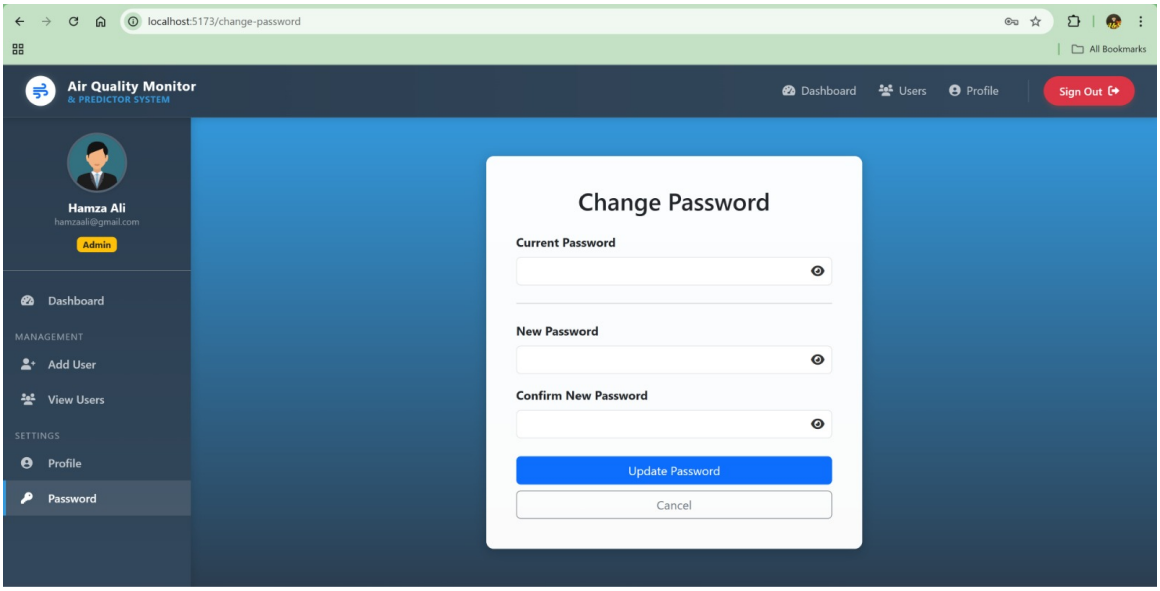


Figure 5.13: Admin Change Password Screen

5.2 User Tables

Custom table types are personalize, temporary data storage structures craft by users using the schema definition. Users can give these table types specific features like primary keys, unique constraints, default values, and other characteristics as need. These tables are essential in the proposed system as they enable the identification of users, their registration, and maintenance. By using the tables, it becomes possible to offer users personalization features and data. In the proposed system, the database is used to record every user's achievements in taking advantage of surveying tools and solutions. In the given system, the database is an integral element because it stores all data about the achieved progress in learning every particular aspect related to surveying tools and solutions.Every milestone made by the users takes part in supporting them to continue exploring and applying the survey equipment and solutions. [24].

The reference table contains the information about the parameters of the air quality that can help to monitor and predict its behavior in an informative and organized manner. Such a presentation allows a better visual understanding, sorting, and comparing of the data. Therefore, the table is a valuable source of information on air quality sensors. Table 5.1 undefined

Table 5.1: User

Field Name	Description	Type	Length	Default	Not Null
Email address	User's Email address	String	40	identify	Y
Password	User's Password	String	15		Y

Keys		
Table Name	**Fieldname**	**Key Type**
tbluser	Email	Unique
tbluser	Password	Unique

Table 5.2 The signup table contains information related to users' signups, such as their username, email, password, and user roles. It validates the users input, ensuring that the data is safe and appropriate, and allowing the user to gain access to the application's features and provide them with an individual experience.

Table 5.2: Sign-Up

Field Name	Description	Type	Length	Default	Not Null
User Name	User's Name	String	30	identify	Y
Email Address	User's Email Address	String	30		Y
Password	User's Password	String	15		Y

Keys		
Table Name	**Fieldname**	**Key Type**
tblSign-Up	Email	Unique
tblSign-Up	Password	Unique

Table 5.3 The user management table provides an administrator with the ability to view, add, change, or delete users' data. The table stores the information about users' roles, status, and access rights, which helps manage and control the users' activity efficiently.

Table 5.3: User Management

Field Name	Description	Type	Length	Default	Not Null
User Name	User's Email	String	30	identify	Y
User Profile View	User's Action	Boolean	5		Y
User Profile Delete	User's Action	Boolean	5		Y

Keys		
Table Name	**Fieldname**	**Key Type**
tblAdmin	User Profile View	Primary
tblAdmin	User Profile Delete	Primary

Table 5.4 The air quality predictor table stores historical sensor readings and prediction results. It organizes pollutant levels, timestamps, and forecast values, enabling accurate analysis, trend evaluation, and efficient retrieval of predictive data for monitoring air quality conditions.

Table 5.4: Air Quality Predictor

Field Name	Description	Type	Length	Default	Not Null
Temperature	Temperature	Decimal128	4		Y
Humidity	Humidity	Decimal128	4		Y

Continued on next page

Table 5.4 – *Continued from previous page*

Field Name	Description	Type	Length	Default	Not Null
CO2 Level	CO2 Level	Decimal128	8		Y
Air Quality	Air Quality	Decimal128	8		Y

Keys		
Table Name	**Fieldname**	**Key Type**
tblAir Quality Predictor	Temperature	Primary
tblAir Quality Predictor	Humidity	Primary
tblAir Quality Predictor	CO2 Level	Primary
tblAir Quality Predictor	Air Quality	Primary

Table 5.5 The password table stores user authentication credentials securely, including hashed passwords and associated user IDs. It validates login attempts, enforces security policies, and supports password updates or resets, ensuring protected access and maintaining the integrity of the system's authentication process.

Table 5.5: Change Password

Field Name	Description	Type	Length	Default	Not Null
Password	Current Password	String	15	identify	Y
New Password	New Password	String	15		Y
Confirm Password	Confirm Password	String	15		Y

Keys		
Table Name	**Fieldname**	**Key Type**
tblChange Password	Password	Unique
tblChange Password	New Password	Unique
tblChange Password	Confirm Password	Unique

Chapter 6

Test Plan

The chapter is, about the systems validation process. The systems validation process uses test strategies to see if the core components are working correctly if they are stable and if they are reliable. The chapter also talks about the things that are needed to test the system like the testing environment and the tools. It explains how to come up with test cases and how to figure out if the system is working the way it should in situations. Each part of the chapter goes into detail about the testing phases and makes sure the system is safe and works the same every time the systems validation process is very important. [25].

These parts talk about the software testing environment and the tools used for software testing. They also talk about what to achieve with software testing. When the software testing is done. The test plan also says who is responsible for what and what to do to get ready for software testing.

6.1 Unit Testing

Unit testing is an essential process that allows checking the functioning of particular segments of code in the air quality monitoring and prediction system. The segment can be a function, a procedural module, or a particular service, for instance, sensor data acquisition, data filtering, calculation of the air quality index, or prediction algorithm. The process under consideration isolates each segment and checks its functions to ensure that it operates correctly in isolation. Testing should verify that all the segments, including sensor calibration, data validation, API interaction, prediction results producing, and others, work properly under specific conditions. The tests can check if the sensor works correctly, the data is acquired and filtered correctly, if the prediction results are relevant. The testing ensures that interfaces, connections, and all other aspects of the application work correctly. Through automated unit testing, segments can be repeatedly tested after any changes to detect breakdowns, fix issues, and verify the correctness of the implemented functions. Tests can serve as documentation that describes how the segments under consideration work, which contributes to easier maintenance, system improvement, and development. The unit testing should ensure the continuous validation of the system under consideration, thus helping to achieve the high performance of the air quality monitoring and prediction system. [26].

[Table 6.1]By offering continuous verification, unit testing contributes to the creation of an efficient, reliable, and scalable air quality surveillance and prediction system, as it helps each module perform its intended duty consistently and adapt to emerging technological needs throughout the development cycle. Table 6.1 undefined

Table 6.1: Unit Test casr for Authentication

Test Case	Test Objective	Test Steps	Test Data	Expected Results	Pass/Fail
UT-01	01	1. Sign Up	To verify new account registration with valid and invalid data.	Successful account creation with confirmation message.	Pass
UT-02	02	1. Sign In	To validate user authentication using correct and incorrect credentials.	Access to dashboard after valid credentials.	Pass
UT-03	03	1. View Profile Information	To confirm accurate retrieval and display of user details.	Profile information displayed correctly from database.	Pass
UT-04	04	1. Update Profile	To test updating user information and database reflection.	Profile updated and confirmation message displayed.	Pass

6.2 Integration Testing

Integration testing ensure that all parts of the proposed air quality monitoring and prediction system work together as intended. Testing show that air quality sensors adequately communicate information about particulate matter and carbon dioxide levels as well as other pollutants to the controller. Integration testing communication problems between separate components and ensure that the information on the display is reliable. Issues arise the sensors may provide data in an unacceptable format or with delays, disrupting communication with the controller and prediction module. Integration testing ensure reliable performance of the application and hardware conditions. Check proper communication between the application and hardware, which is critical for timely response to high pollution days. Ensuring that the data provided by sensors is delivered to the right place, integration testing reduces errors and enables faster and more accurate prediction. [27].

[Table 6.2]Integration testing ensures that the interconnected modules function together correctly, maintaining data flow, synchronization, and transaction accuracy across different components, enhancing system reliability, security and performance. Table 6.2 undefined

Table 6.2: Integration Testing for Air Quality Predictor

Test Case	Test Objective	Test Steps	Test Data	Expected Results	Pass/Fail
IT-01	01	1. Sensor Reading	To verify proper interaction between sensor reading interface, database.	Relevant results retrieved and displayed accurately.	Pass
IT-02	02	1. KNN Model	To verify the test case model run according to the data provided.	Shows the predicted data on screen successfully.	Pass
IT-03	03	1. Solidvector Model	To verify the test case model run according to the data provided.	Shows the predicted data on screen successfully.	Pass

6.3 System Testing

System testing is designed to ensure that the air quality monitoring and prediction system operates effectively under the intended conditions. The process is used to assess the precision of its operation, performance, reliability, and security. Through the use of system testing, one can be assured of the proper functioning of all the features and components, which include the sensors, the data acquisition and processing modules, prediction models, and alerting mechanisms. The test also assures that the system processes data continuously and accurately, provides pollution predictions, and facilitates timely alerts. It determines the performance of the application under heavy data traffic, such as when multiple sensors are transmitting data at once, and its ability to maintain regular operations when subjected to elevated loads. The system test determines the system's ability to maintain its functionality and the integrity of the data being transmitted in case of network outages. Testing ensures the most efficient and reliable performance of the system, thereby facilitating its effective use by both users and administrators. These enhances the level of trust that the former has in the latter while interacting with the system. [28].

[Table 6.3]Explores end-to-end system workflows, ensuring smooth integration of modules such as sensor data collection, air quality analysis, prediction output, alert generation, and user interaction. The evaluation confirms consistent and accurate operation, enhancing reliability, system performance, data accuracy, and secure, seamless monitoring under normal and high-load conditions. Table 6.3 undefined

Table 6.3: System testing for validation

Test Case	Test Objective	Test Steps	Test Data	Expected Results	Pass/Fail
ST-01	01	1. Prediction Report	To verify proper interaction of prediction data on provided data.	Successfully display prediction report accurately.	Pass

Continued on next page

Table 6.3 – continued from previous page

Test Case	Test Objective	Test Steps	Test Data	Expected Results	Pass/Fail
ST-02	02	1. Feedback	Check end-to-end flow: submission, storage, and admin review of feedback.	Feedback submitted by user is stored and visible to admin for review.	Pass
ST-03	03	1. Logout	Ensure session termination, token revocation, and redirection.	User is logged out completely, tokens invalidated, and redirected to login page.	Pass

Chapter 7

Conclusion and Future Work

The system is designed to monitor air quality on an ongoing basis. It also helps to predict air quality levels by analyzing the data it collects. The system is reliable and ensures that the data it collects is safe and securely transmitted. The air quality monitoring system is reliable and has been proven to be stable under various environmental conditions. The next step in the proposal would be to improve the accuracy of the predictions by developing better predictive models by employing machine learning algorithms. Other features such as notifications and visualization can be added to provide an enhanced experience for the user to monitor air quality in various environments.

7.1 Conclusion

The air environment monitoring and prediction system has good scalability and can effectively ensure the security of monitoring information. The air environment monitoring and prediction system is powerful in handling business processes and has excellent performance in processing massive amounts of data. The system uses sensors, data analysis modules, and prediction modules to work together to ensure the accuracy of data. It improves the performance and stability of the system through the combination of various modules. The system has a high degree of flexibility, which can meet the requirements of diversified and personalized scenarios and have important significance for diversified and personalized scenarios, Air environment monitoring and prediction system.

7.2 Future Work

The air quality monitoring and prediction system will be improved by developing the accuracy of the forecasts with the help of additional machine learning algorithms. All the sensors will be connected to provide the most extensive coverage possible and to collect more data, such as the level of humidity and temperature. The data will be stored on cloud servers. The system will also offer mobile and web applications that would make the dissemination of information more convenient and attractive. The system will be improved by providing better security and privacy features for its users.

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